

The Lingua Handbook

Everything Lingua, from start to finish

DESIGN · PRODUCT · PEOPLE · LAUNCH · RUNNING IT

Third edition, October 2026 (updated 28 October)

Learn a language
by living in it.

LINGUA

This book is for whoever works on Lingua next: you, a friend, or a Claude on another account. It says what Lingua is like, what it does, who it is for, how to tell people about it and how to run it, so that every page and every step can become a little more like it.

It lives in the repository at `docs/handbook`. Print it, scribble in it, and change it when Lingua changes.

BEFORE YOU BEGIN

How to use this book

Read Part One once. It is short, and everything else follows from it.

When you sit down to design a page, open its chapter in Part Four. Every room fits on one page: what the room is for, what it shows first, what it keeps quiet, the one thing to do there, and three questions to ask yourself.

The exact things live in the margins. Colours, sizes, file names, springs and dates are written small, beside the sentence they belong to. You can read the book without them, and you can build from them without reading the book.

LIKE THIS

The exact things live out here, small, beside the sentence they belong to.

Before you call a page done, go through Part Five. Its review sheet is meant to be printed and ticked, one sheet per page.

Parts Two and Six to Nine are the rest of the work: what Lingua does and where it is going, who it is for, telling people, running it, and the whole road from here to version 1.0.

At the back are every number in one place, the Latin names and what they mean, Lingua 7 at a glance, and a page for your own notes.

A few marks

in the margin	the fine print: exact values and where they live
~~*struck through*~~	something Lingua had and took out, kept so we remember why
☐	something to check
	a pause

THE WHOLE BOOK

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PART ONE

The spirit

What Lingua is, how it behaves, and how it talks.



CHAPTER 1

What Lingua is

Lingua is a place to live in a language for a while. You read in it, listen to it, watch it, talk and write in it, and every word you meet is kept, so it comes back just before you would have forgotten it.

It is not a game and it is not a school. It is closer to a study at home: a desk by the window, a shelf of books, a pot of tea going cold because the chapter got good. That is why its design is called the Cozy Scriptorium.

The rooms have Latin names, because a scriptorium would have used them, and because it is a little bit fun. Each room keeps one kind of work:

Home	your hearth: today, and what's next
Cursus	the course, grammar, alphabets and typing
Codex	reading: books, articles, comics
Thesaurus	every word you keep, and Ludus, the games
Sonus et Visio	listening and watching
Mentis	the guide, talking and writing
Ratio	how it's going
Settings	everything you can change

Lingua is made for the desktop. It uses a wide window the way you use a desk: things side by side, nothing stacked that could sit next to each other.

THE TOKENS

`src/design/tokens.css`, "Cozy Scriptorium design tokens".

ORDER

`src/app/sections.ts`. Alt 1 to 8 opens them in this order. Ratio and Settings sit apart at the bottom of the sidebar.

DESKTOP ONLY

Since 2026-10-20. The user: "Phone is not needed anymore." What exists for the phone stays; nothing new is made for it.

CHAPTER 2

Six habits of a Lingua page

Six habits, more than rules. A page that has all six usually needs nothing else.

I. The language comes first

On every page, the language is the biggest, best-set thing there is. A sentence in German gets the serif, the space and the light; the buttons around it step back. If a page has nothing in the language on it, ask whether it should.

SET IN

Titles in Literata. Reading text in the learner's own reading font, one of six.

II. Quiet until asked

Show the one thing that matters and keep the rest a click away: details in a drawer, numbers on hover, options under More. Control is never taken away, only folded. Whoever wants every setting finds it; whoever doesn't never has to see it.

BUT

Nothing important lives only on hover. Whatever a hover shows, a click or a key shows too.

III. One word, everywhere

A word saved while reading is the same word in Thesaurus, in Ludus, in a talk in Sermo and in Ratio's counts. Design as if each word were a small thing the learner carries from room to room, and let it remember where it was found.

WHERE WORDS LIVE

The word memory and the dictionary, Thesaurus › Dictionary (4.0-2).



IV. Honest, never hopeful

TAKEN OUT

2026-10-26: the first start's "B1 in about 7 months" and its pace verdicts. Not accurate enough to say.

Say only what is true and built. No forecasts Lingua can't keep, no scores that look precise and aren't, no features promised before they exist. When Lingua doesn't know, it says so plainly, or says nothing at all.

V. The learner holds the pen

BORROWED

From the Lingyu blueprint (2026-10-25): "saving is not scheduling"; anything generated is editable and can be dismissed.

Whatever Lingua decides for you can be undone: a word marked known can be unmarked, a level changed, a card edited. Nothing large happens behind the learner's back. Saving a word is not the same as scheduling it, and one click never quietly grows a deck of fifty.

VI. Wide and calm

SIZES

Check at 1280, 1440 and 1366 × 768. Shortcuts: Alt 1–8 for the rooms; the rest in the Shortcuts dialog.

Use the width: columns, side panels, sticky rails, a list beside its detail instead of a page after it. Every page fits a 1366 × 768 laptop with no little scroll bar sneaking in. The reader and long lists are the only things that scroll, and they do it on purpose.

If you only remember one: the language first, everything else quiet.



CHAPTER 3

The voice

Lingua talks like a friend who reads a lot: informal, warm, a little dry. It likes to make you smile, but never at the cost of being clear.

One smile per page. A funny subtitle is a gift; three of them are a stand-up set. The joke goes where nobody has to read it to act: a subtitle, a quiet line under a field. Never inside an instruction.

Titles ask or name. "What language will it be?" "So, how far along are you?" "Choose your study light." Short enough to take in at a glance, and a real question when there is one.

Buttons say what happens. "Continue with Deutsch" says more than "Next". One main button per page.

Plain where it matters. Money, privacy, errors and deleting get the plainest words in the app. "Could not save your choices. Please try again."

Small manners. British spelling, as in colour and favourite. Curly quotes, and the language's own quotation marks when its touches are on. No exclamation marks in the interface; the welcome toast may keep its one.



THE USER'S OUTLINE

For the first start: "minimal, informal, bookish, building up the excitement, with funny subtitles" (5.1–2b).

REAL LINES

From `Onboarding.tsx`: the error above, and the welcome toast, "Welcome to Lingua, {name}. Happy reading!"

MARKS

« French » · „German“ · 「Japanese」

Said, and rather not

Lingua's own lines on the left. The struck-through ones were in Lingua and were taken out.

WE SAY	WE'D RATHER NOT
What language will it be?	Select your target language
Continue with Deutsch	Next
8 of 15 minutes	53% of your daily goal complete!
It stays on this device. We just like to be polite.	We value your privacy.
Click a country to see what's spoken there, or scroll down for all of them.	~~… Commitment issues welcome. you can add more later.~~ A joke inside an instruction.
(nothing, when Lingua can't know)	~~At 15 minutes a day, B1 in about 7 months.~~ A forecast it couldn't keep.
Your shelf is empty. Import a book, or take one from the library.	No items found.
None of Lingua's languages is spoken in {country} yet.	No results.

FOR EXAMPLE

"Your shelf is empty …" is an example of the voice, not a line in the app.

Read your page aloud in the voice of someone who likes you. If a sentence would sound strange said to a friend across a table, change it.



PART TWO

The product

What Lingua does today, in which languages, for what price, and where it is going.



CHAPTER 4

What Lingua does today

SOURCE

`docs/FEATURES.md`, kept up to date with every step.

Room by room, what a learner can do in Lingua now (version 0.5.2, released 2026-10-28). The details are in FEATURES.md; this is the shelf, not the books.

ROOM	WHAT IS THERE
The map	Since 0.5.2-alpha.17 the main rooms are Home, Cursus, Explora and Bibliotheca; Codex, Thesaurus, Sonus, Visio and Mentis sit under Tools; Ratio and Settings are at the bottom. Explora is one place to discover (reading, films, series, radio, audiobooks, words) at your level, in three bands: at your level, a stretch, easier. Bibliotheca is the one library across every tool.
Home	Good evening by name; today's minutes against the goal on a ribbon by room; Next up, one click each; the week; first steps for new learners; the next grammar topic and Mundus's suggestion. A Classic layout a setting away.
Cursus	The course for the six core languages: twelve units A1–B1 as an adventure trail, each with words, phrases, grammar, speaking, writing and a check. Grammatica (topics A1–B2 for the core six, short grammars for twelve more), Alphabets for seven scripts with writing, and keybr-style typing in the language's layout.
Codex	Read anything: EPUB, PDF, text, comics with speech bubbles read on the device, photos of printed pages, news, Wikipedia, Gutenberg and Wikisource, Goodreads shelves. Every word a click away, phrases by dragging, sentences translated ahead, the word panel, typography and papers, the end page, Check yourself.
Thesaurus and Ludus	Decks in folders, FSRS scheduling, 47 ready-made decks, Anki and CSV import, the offline dictionary for the core six (forms, recordings, etymology, Tatoeba), the word memory with every place a word was met. Ludus: choices, pairs, gaps, word banks, typing, speaking, letter tiles, listening and dictation, Lightning.

ROOM	WHAT IS THERE
Sonus et Visio	The library (songs, podcasts, episodes, audiobooks, stations, your files, films and videos); LibriVox and free courses; radio in the language with a tuner and live text; Sonus+ for music with timed lyrics, song lessons and podcasts; Visio for films, series and YouTube links with two interactive subtitle lines, a follow clock for a show playing elsewhere, and a browser extension for the services' subtitles; Commute mode.
Mentis	Mundus, the guide: a note on your week, three next steps that answer back (something shorter, I'd rather listen, why this, not today; with Max, your own words), and Things to try, small cards with a time chip, a tick that stamps them, filters, and Deal me one. Sermo: about 80 scenes in groups, each ending in a prompt for your own AI app (ChatGPT, Claude, Gemini, Copilot, Grok); the chat inside Lingua is gone. Diarium, the journal, with writing tools, a calendar, search, checking by AI with Max, a word diff and corrections to cards.
Ratio	Overview for the language being learned: an estimated level, words, reading, listening, cards, writing, typing, grammar and a year of habit; Tabula, every language you speak, with goals; Log time.
Settings	Eight groups with search: reading, cards, listening, languages, look and feel, AI and translation, your data, about.

Across the whole app

- Works offline as an installed desktop app, and updates itself at a quiet moment.
- Your devices sync directly, device to device, with no account and no server; backups as files.
- Natural voices: real speakers' recordings, Piper voices on the device, Gemini with a key.
- Paper and Nocturne, twelve palettes, six reading fonts, a touch of each language.
- Keyboard first: Alt 1–8 for the rooms, letters for words, a shortcut sheet.

Next: Lingua 6's features, then Lingua 7's design pass.

What changed lately

Since the last edition of this book (versions 0.5.2-alpha.17 to 0.5.2, late October 2026). Each line is a step of Lingua 5.2; the details are in `docs/FEATURES.md`.

VERSION	ROOM	WHAT A LEARNER NOW HAS
alpha.17	*The map*	Home, Cursus, Explora and Bibliotheca are the main rooms; Codex, Thesaurus, Sonus, Visio and Mentis sit under Tools; Ratio and Settings at the bottom.
alpha.18	*Diarium*	A real editor (headings, lists, a toolbar over a selection), “Cher journal” in the language, search and sort, rename and delete with Undo, a calendar of the days written.
alpha.21	*Sermo*	Three screens that build a prompt and open the learner’s own AI (ChatGPT, Claude, Gemini, Copilot, Grok) with it; about eighty scenes. The chat inside Lingua is gone; old chats stay and open Sermo.
alpha.22–.24	*Mundus*	Next steps that answer back (shorter, I’d rather listen, why this, not today); one Explore; Things to try as small cards with a time chip, a stamp and Deal me one.
alpha.25	*This book*	Money, what Lingua could earn, and how to fund it.
alpha.26	*Grammatica*	A guided path at the learner’s level with the next step and its reason; a ladder of levels; lessons that use the width of a desktop window.
alpha.27	*Ratio*	One language at a time; Iter retired and its goals moved into Tabula with “on track” and “behind”; an estimated level said as an estimate.
alpha.28	*Ratio, Visio*	Look-ups per 1,000 words, listening by kind, a typing card, Log time and Add a moment; YouTube in Visio, the follow clock, and the Lingua extension (not yet tried on real sites).
alpha.29	*This book*	The design chapters: the desktop frame, layout patterns, elements, where things go, how it feels, dark, charts, words, a recipe; tablet and phone; a layout page for every room.
alpha.30 to 0.5.2	*This book*	Icons, forms, charts, accessibility, sound, the first start, a cheat sheet. Lingua 5 released as 0.5.2.

SOURCE

`docs/ROADMAP.md` › Finishing Lingua 5.

Lingua 5 is complete (0.5.2, 2026-10-28). The check was a sample; `ROADMAP.md` 5.2-10 lists its gaps.

The design changelog

What changed in how Lingua looks, sounds and is built to be designed, by version. The product's own changes are on the page before; this one is for people drawing screens. **Keeping it up:** when a push changes a token, a primitive, a pattern or a rule here, add a line to this table and, if a number moved, to the facts block; `docs/HANDOVER.md` says so.

VERSION	PART	WHAT CHANGED
0.5.2	*Release*	Lingua 5 released, 2026-10-28. No token or primitive changed; the book was reprinted with its margins checked.
alpha.31	*Rules and chapters*	Forms and validation, the anatomy of a chart, the fourteen papers measured, a screen in Nocturne, the design brief, accessibility, sound and haptics, the first start screen by screen, empty-page art, a cheat sheet checked by a test. Two findings: terracotta and sage share one lightness; faint ink fails 4.5 : 1 on Parchment and Mocha.
alpha.30	*Icons*	The Lingua hand: 33 icons on a 48 grid, one fine line and one terracotta detail, as files and as pages. Not in the app yet.
alpha.29	*Design chapters*	The desktop frame, patterns, elements, where things go, how it feels, dark, charts, words, a recipe; a layout page for every room; tablet and phone as guidance.
alpha.28	*Ratio, Visio*	A typing card and by-kind bars in Ratio; Log study time and Add a moment as drawers; the follow clock in Visio.
alpha.26–27	*Grammatica, Ratio*	Full-width lessons and a ladder of levels; one language at a time with goals in Tabula.

The rule for what goes in

- **In:** a new or changed token, primitive, pattern, icon, sound, or a rule that a designer must now follow; a measured finding.
- **Out:** a feature that used the existing pieces as they are. That belongs in the product table.
- **One line** a version, with the effect first. If a line needs a paragraph, the paragraph belongs in the right chapter and the line points to it.

CHAPTER 5

Languages and tiers

WHERE

`TIERS` and `LANGUAGES` in `src/core/lang/languages.ts`.

Seventy languages, in five tiers that say honestly how much Lingua has for each. A tier is a promise: nothing on the page may claim more.

TIER	LANGUAGES	WHAT THEY GET
Core	6	Everything: the course, grammar, the alphabet, typing, the full dictionary, natural voices, every room
Full	14	Every room, grammar and the alphabet; no course
Beta	20	What builds itself: the reader, a dictionary, cards and Ludus, podcasts, radio, music search
Basics	20	Reading with tap-to-translate, your own words and decks, radio and podcasts
Experimental	10	Reading with machine translation where it exists

RULE

Tier 2 stays cheap: only features that cost little to keep (automatic packs, decks, podcasts, music search). CLAUDE.md.

The core six are Spanish, French, German, Russian, Arabic and Mandarin. Languages not built yet are shown with Soon and can't be chosen.

For marketing: say "six languages with everything, and sixty-four more to read and listen in". Never "70 languages" alone: it sounds like seventy courses.

CHAPTER 6

Adding a language

A language climbs the tiers one step at a time, and each step is real work. Never list a language higher than what is built and checked for it.

HOW

`docs/SETUP.md`, section 5, has every command. Pack builds take 1–2 GB of memory each: one at a time, dev server stopped.

TO REACH	WHAT IT TAKES
Experimental	An entry in `LANGUAGES`: code, name, endonym, script, flags, a sample line. Reading with machine translation where it exists.
Basics	Its flag or emblem in `design/flags.tsx`; radio stations and podcasts found in the language.
Beta	A language pack from Wiktionary (kaikki) and FrequencyWords: `fetch.ts`, `count.ts` where needed, `build.ts`. Then decks, Ludus and music search come by themselves.
Full	Grammatica topics, each checked against a named source; the alphabet if it has its own script; Mundus's picks chosen by hand.
Core	The full dictionary (`dictionary-fetch.ts`, `dictionary.ts`), recordings (`recordings.ts`), typing (`typing/build.ts`), the twelve-unit course, a language touch, a Piper voice where one exists, and a native reader's check (next chapter).

Tier 2 stays cheap: below Core, only what builds itself or costs little to keep. A new Core language is a Linguasized step of its own, planned like one.

RAISING A PACK

A higher `PACK_VERSION` makes installed packs refresh quietly.

CHAPTER 7

Getting the language right

A learner believes every sentence Lingua shows them. A language app is only as good as its language.

ALREADY DONE
The commonest words' meanings hand-checked in Spanish, French, German and Russian (5.1–3a).

Name the source. Every grammar rule is checked against a named source, listed under it, as Grammatica already does. Data keeps its origin: Wiktionary, Tatoeba, Lingua Libre, each with its licence.

A native reader before Core. Before a language becomes Core, and before the interface is translated into it (Lingua 6-19), someone who speaks it natively reads the course, the grammar and the interface. Pay them or thank them by name on the credits page.

Say which variety. Castilian or Mexican Spanish, European or Brazilian Portuguese: the flag's accent decides the voice, and the words say which they are.

Generated text is a draft. It is labelled, checked before it is shown (Lingua 6-6) and can be reported.

TEST SENTENCES
A fixed handful per language (forms, right to left, Japanese without spaces) read in the reader before each release.

Reports become fixes. A wrong meaning reported by a learner goes on a list, is fixed at the next pack build, and the learner is thanked.

CHAPTER 8

Plans and prices

Free is generous, Lingua+ is bought once, Max is AI with your own key. Never pushy.

	FREE	LINGUA+ · €24.99 ONCE	MAX · €5.99 A MONTH, €49.99 A YEAR
Codex	Everything: reading, look-ups, imports, Discover	—	AI sentence translation, Explain in context
Thesaurus	Unlimited decks and cards, Discover	Ludus	—
Sonus et Visio	Library, audiobooks, radio, your files	Sonus+, streaming and subtitles	Automatic dictation
Cursus	Alphabets	The course, Grammatica, typing	Feedback on writing and speaking
Mentis	Mundus, Diarium	Sermo	Diary checking, natural AI voices
Ratio, Settings	Everything	—	—

Max includes all of Lingua+ while it runs; whoever bought Lingua+ keeps it forever. Max's AI runs on the learner's own free key (Groq or Gemini), so Lingua has no AI server and no AI bill.

The honest bit to explain: Max asks for a monthly price and the learner's own key. Its page must say plainly what Max adds, that the key is free from its provider, and that the provider is separate from Lingua.

DECIDED

D1–D4 and D8 of the 5.1 plan (2026-10-20). Stripe; prices as above.

UNTIL STRIPE IS LIVE

Test mode: buying switches the plan and charges nothing. Installs from before the plans act as Max.

CHAPTER 9

Where Lingua is going

Each Lingua is a plan of its own, built step by step, each step pushed with a version.



QUEUE

`docs/HANDOVER.md`, steps 17, 18, 22 and 21.

Lingua 5.2 has every step built: Radio and listening, Diarium, Sermo, Mundus, Grammatica's guided path, Ratio and Visio. What is left is its check, which makes version 0.5.2.

PLAN

`docs/LINGUA-6-PLAN.md` and the audit `LINGUA-6-FEATURES.md`; step 23. Versions 0.6.x.

Lingua 6 · Features comes next (your call, 26 October): it adds what the Lingyu catalogue has and Lingua doesn't. It is staged so the parts that can run side by side go to different AIs (Claude, ChatGPT, Gemini), after step 6-0 writes the shared types, tables and APIs as typechecked code. The next chapter is its summary.

PLAN

`docs/LINGUA-7-PLAN.md`; step 24. Versions 0.7.x.

Lingua 7 · Design follows, room by room from your notes, starting with the first start. This book is its guide.

VERSIONS

`docs/VERSIONS.md`.

1.0 is the first public release. The version leaves alpha (0.x) only when you say so.

CHAPTER 10

Lingua 6 · Features

On 26 October 2026 every entry of the Lingyu catalogue, all 1,618, was checked against Lingua. Most of the core is there; the gaps are deep rather than wide.

THE AUDIT

`docs/LINGUA-6-FEATURES.md`: each entry ticked or given the step that builds it.



437 there · 262 partly · 834 missing · 38 in 5.2 · 47 to decide

Foundations first

Many entries are one thing seen from several rooms: synonym cards, synonym races, synonym maps and a synonym finder while writing all come from one source. So Lingua 6 builds six shared pieces first, and the rooms grow on them. That keeps 1,143 entries from becoming 1,143 separate features.

PART A

The word record · the inbox and decks · cards and memory · one exercise engine · the learner's record · generated materials, checked.

The same rules as always

- Every feature lands in a room that exists; new tabs only where a room needs one.
- Local first; AI only with Max and the learner's key; estimates labelled.
- Hand-made content for the core six; what builds itself for the rest.
- Everything generated is checked before it is shown, and labelled.
- Saving is not scheduling: previews before a deck grows.

The 28 steps, in seven parts

A · FOUNDATIONS

6-1 The word record
6-2 The inbox and the deck workshop
6-3 Cards and memory
6-4 One exercise engine
6-5 The learner's record
6-6 Generated materials, checked

B · THE SKILLS

6-7 Grammar
6-8 Reading
6-9 Listening
6-10 Sounds and pronunciation
6-11 Speaking
6-12 Writing
6-13 Translating and comparing

C · PATHS AND GUIDANCE

6-14 Courses and lessons
6-15 The tutor, in context
6-16 Personal and adaptive
6-17 Goals, habits and progress
6-18 Exams and levels

D · MORE OF THE LANGUAGE

6-19 Immersion and culture
6-20 Phrasebook, notebook, word map
6-21 Games and projects
6-22 Dictionary tools
6-23 Tools for each language

E · AROUND THE APP

6-24 Import, export, integrations
6-25 Accessibility and the desktop

F · WITH YOUR WORD ONLY

6-26 Experiments
6-27 Together, and beyond the desk

G

6-28 The check; version 0.6.0

Your answers (26 October 2026)

Q1	Which first, Lingua 7 or Lingua 6?	**Lingua 6 first**, then Lingua 7
Q2	Sharing by files and links, or a server with accounts?	**Still to be considered**
Q3	A live voice conversation inside Lingua again?	**No**
Q4	The interface in the language you learn?	**Yes**
Q5	Streaks optional, stamps not points, reminders off by default?	**Yes**
Q6	Phone apps and widgets still decided against?	**Can be**: not ruled out

Q2 and Q6 stay open: nothing in step 6-27 is built until you schedule it.

PART THREE

The materials

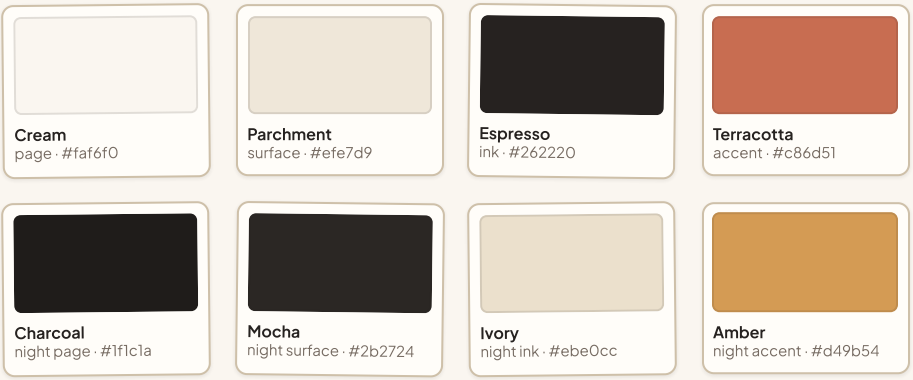
Paper, ink, type, space, motion, and the little drawings.



CHAPTER 11

Paper and ink

Lingua's light is called Paper and its dark is Nocturne. Neither is the other turned inside out: Paper is cream with espresso ink, Nocturne a warm charcoal with ivory, so nothing is ever pure white or pure black.



TEXT IN THE ACCENT

Use `--accent-text` (#a5533b), which keeps 4.5 : 1 on cream. `--accent` itself is for decoration in Paper.

SAGE

#6b8e78; its text #4f6f5b. At night #93b39e.

INVERSE

#3b302a on #f8efe0; at night #e8d8bb with #2b231e.

The accent is a bookmark, not a highlighter. One terracotta thing per area: the chosen language, the next step, the new word. If two things on one card are terracotta, one of them is wrong.

Sage means known, done or growing: a ticked step, a word you know, a finished chapter.

Tooltips, toasts and main buttons are inverse: espresso with cream by day, old-paper cream with espresso at night. Never black, never white.

Palettes and reading papers

The learner picks a palette for each light, and a paper for reading in Codex. Design in Cream and Charcoal, then look at one more of each before you call it done. A page that only works in Cream isn't done.

PAPER



NOCTURNE



Reading papers, by day: Cream, Ivory, Parchment, Sepia, Sage, Blush and Stone. By night: Charcoal, Espresso, Sepia night, Ink, Forest, Plum and Midnight. The paper turns with the light, in the same transition.

Use the tokens, never a colour by hand. A palette swaps the page, the surfaces and the accent. A colour typed in by hand is right in one palette and wrong in eleven.

WHERE

`src/design/palettes.ts`, `src/design/papers.ts`. The first of each list is the original look.

CONTRAST

Each palette's accent text is chosen to keep 4.5 : 1 on its own page. Ink keeps the theme's own.

TOKENS

`--bg --surface --raised --sunken --ink --ink-soft --ink-faint --line --line-strong --accent --accent-text --accent-fill --sage`

CHAPTER 12

Type

Two voices. Literata is the book: the titles, the language, and the numbers that matter. Plus Jakarta Sans is the helper: labels, buttons, explanations. When in doubt, ask who is speaking.

LITERATA

Am Abend lasen wir am Fenster, während draußen leise Schnee fiel.

PLUS JAKARTA SANS

Continue with Deutsch · Click a country to see what's spoken there.

WHERE	FACE	SIZE
Big moments (your name, the welcome)	Literata 600	40–44 px
A page's title	Literata 600, tight	30–36 px
A line in the language	Literata	19 px
Body and buttons	Jakarta	15 px
Small lines under things	Jakarta	12.5 px
Labels, in capitals	Jakarta 700, +0.08 em	11.5–12 px

WHERE

`--font-serif`, `--font-sans` in `src/styles/index.css`; reading fonts in `src/design/fonts.ts`.

The learner picks the reading font, so design for all six: Literata, Lora, Newsreader, Merriweather, Atkinson Hyperlegible and Comfortaa. Scripts they don't cover fall back to Noto Serif SC, JP and KR, Noto Naskh Arabic, or the device's own faces.

Numbers in a table line up (tabular). Numbers in a sentence are the book's own, set in the serif: "8 of 15 minutes".

CHAPTER 13

Space, corners and light

A Lingua page breathes like a well-set book: generous margins, one clear column, and things grouped by space before they are grouped by boxes.

Lines before shadows. A hairline separates; a shadow lifts. Most things only need the line. When something does float, one shadow is enough, and it is warm, never grey.

LIGHT

elev-1 a card at rest · **elev-2** something floating: the map, a popover · **elev-3** dialogs and drawers. All tinted espresso.



Round, not bubbly. Pills for small choices, generous corners for cards and dialogs, gentler ones inside them. An inner corner is always smaller than the one around it.

TAILWIND

Buttons `rounded-lg/xl/2xl` by size; cards and dialogs `rounded-3xl`; empty states `rounded-2xl`.

One screen. A desktop page fits 1366 × 768 without a scroll bar. If it doesn't fit, take something away before making something smaller.

SINCE

2026-10-25, `5c044f9`: desktop screens on one screen, no tiny scroll.

Columns over stacks. A sticky rail for filters, a panel beside a list, two cards side by side that end on the same line.

CHAPTER 14

Motion

Things in Lingua settle like paper: they move a little, then they are still. Motion says where something came from. It never performs.

SPRING	FOR	STIFFNESS · DAMPING
snappy	pills, toggles, indicators	560 · 42
soft	cards and page content	320 · 34
sheet	drawers and sheets	400 · 40
gentle	the first start's pages	200 · 28
fluid	full motion: pages, reordering	210 · 26

WHERE

`src/design/motion.ts`; the rules in `docs/PERFORMANCE.md`.

FULL MOTION

A setting: pages rise 18 px and settle on *fluid*.

Only transform and opacity move. Never a width, a height or a top: the reader has to stay at sixty frames a second.

The first start turns its pages, 48 pixels sideways with a six-degree tilt. Elsewhere, pages rise 8 pixels as they appear.

Reduced motion means none, not less.

Nothing is slower than the learner. If they click again before something has finished moving, the second click wins.



not this slow

CHAPTER 15

Drawings, icons and flags

Drawings are one fine line in the colour of soft ink, with one small thing in terracotta: a bookmark, a tick, a wisp of sound. They are small, and they never explain what the words already say.

THE LINE

Stroke 1.3, round caps and joins, `currentColor` in `ink-soft`. See the open book and the three vignettes in `IntroPages.tsx`.



Icons are Lucide, as lines. A room keeps its icon wherever it appears: the sidebar, Home, a link from another room.

Flags are drawn, all the same size and corner. Where one country's flag would be wrong for a language, it gets its own emblem (the Senyera for Catalan, the Ikurriña for Basque, the Frisian flag) or a badge with its script on soft paper: ਪੰ for Punjabi, Š / Ћ for Serbo-Croatian, सं for Sanskrit, ᚱ for Old Norse.

An empty page may have a drawing. A full page doesn't need one.

THIS BOOK

The little room drawings here are for the book only. In the app, the rooms use their Lucide icons.

WHERE

`src/design/flags.tsx`. Script badges set the glyph in the reading font on a paper square, so they sit calmly beside real flags.

A touch of the language

WHERE

`src/design/flavours.ts`, on by default: Settings › Look and feel › Language touches.

When you learn French, Lingua looks a little French. Each core language lends the app a heading face, one ornament and a lean of the accent. It is a touch, not a costume: the reading text and the learner's palette never change.

LANGUAGE	HEADINGS	ORNAMENT, AND MORE
French	Garamond	a fleuron, a hint of bleu de France
Spanish	Alegreya	a tile strip; ¿ in titles that ask
German	a clean grotesque	a straight rule
Russian	PT Serif	a thin red rule
Mandarin	a Song face	a vermilion seal for "done"
Arabic	Amiri	an eight-point star

QUOTES

In the language's own marks: « » for French, „ " for German, 「」 for Chinese and Japanese.

Where the touch shows: room titles, Home's greeting, dividers, empty states, the end of a book, the loading screen, and the first start once a language is chosen. Never in the text being read.

When you design a page, switch through the six. If the touch makes anything harder to read, the touch is wrong, not the page.



CHAPTER 16

Icons: the Lingua hand

An icon in Lingua is a small drawing, not a pictogram. It should look as if someone had sketched it with a fine pen in the margin of a book: one line, a little imperfect, and one small thing in terracotta.



The 48 grid: a live area of 40, the circle, the square, the portrait and the landscape that every icon is fitted to.

1. **The field is 48 × 48,** with 4 units of margin all round. Big shapes touch the keylines; small ones centre in them.
2. **One line.** Stroke 2.6 on the 48 field: 1.3 px at 24 px, a little under 1 px at 16, 1.7 at 32. Nothing finer than 3 units apart, or it clogs.
3. **Round caps and joins,** always; no sharp mitres, no square ends.
4. **Drawn, not ruled.** Curves lean by half a unit; a circle is a little egg; lines overshoot a corner by a unit or leave it open.
5. **One idea.** A bookmark is a bookmark, not a bookmark plus a star.
6. **One terracotta detail:** the part that carries the meaning, under a quarter of the line, never filled except for a dot.

Why these, and not a standard set

A standard set is clean and everywhere; Lingua is a book. The pages already draw a little (the books, the sprig), and icons in the same hand make buttons and empty pages feel like one object. The 33 here are the start of that set.

DRAW

- Things from a desk: a pen, a seal, a lamp, a leaf.

AVOID

- Fills, gradients, shadows; compass-perfect circles.

Sizes, and what to leave out

The same drawing is used at every size; what changes is how much of it is shown. Faint detail lines (at half strength) belong to 24 px and up; at 20 and below they are dropped, and the drawing reads from its outline and its one accent.

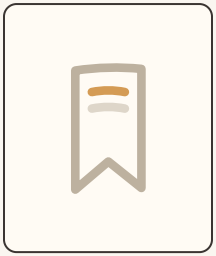


SIZE	USE
16	Tabs, dense lists, chips. Outline and accent only.
20 · 24	Buttons with a label; icon buttons (a 32 to 44 px target around them).
32	Cards and room tiles.
48 · 96	Empty pages, onboarding, the end of a book: here the faint details return and a second accent may be used.

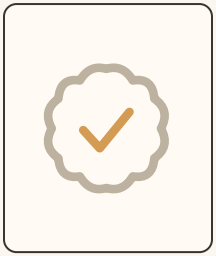
Colour and state

STATE	LINE	THE ONE ACCENT
Rest	Soft ink	Terracotta (amber at night)
Hover	Ink	The same
Selected	Ink, on a soft wash	Terracotta, a little stronger
Disabled	45% of soft ink	45%: it fades with the rest
On an accent button	Ivory (or Charcoal at night)	None: the fill is the accent

At night, and how to draw one



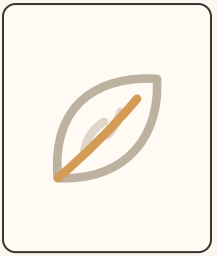
Bookmark
in Nocturne



Seal
in Nocturne



Candle
in Nocturne



Leaf
in Nocturne

At night the line turns soft ivory, terracotta turns amber, sage turns light sage, and a paper fill turns Mocha: the same drawing in a lower light.

How to draw one

1. **Sketch it on paper** with a fine pen, three times, and pick the one that is least careful.
2. **Trace it in a vector editor** on the 48 field, with a 2.6 stroke at 48 (1.3 at 24), round caps, round joins.
3. **Keep few nodes,** twelve at most; let curves do the work.
4. **Pick the one detail** that carries the meaning and make only that terracotta.
5. **Test at 16, 24 and 48,** in Paper and Nocturne, beside two others from the set.
6. **Name it for its meaning** (`bookmark`: a saved word), and write the meaning beside it.

Before an icon is added

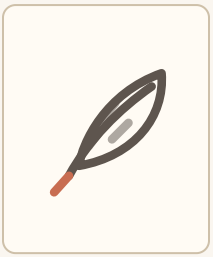
- It means one thing, and that thing has no icon yet.
- It reads at 16 px in Paper and in Nocturne, and beside two others from the set.
- It has an accessible name (its meaning) and, if it stands alone, a tooltip.
- It is saved as `name.svg` in `docs/handbook/icons`, with the line in `currentColor`.
- This chapter's meaning table has a row for it.

Reading and learning

Things from a reader's desk. Each says one thing, and the same thing everywhere.



Bookmark
a saved word



Quill
to write



Open book
a text to read



Stack of books
the library



Letter
the inbox



Reading lamp
focus, night



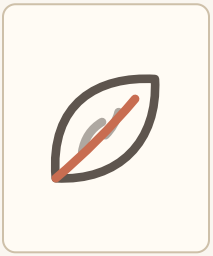
Magnifier
look up



Spectacles
read closely



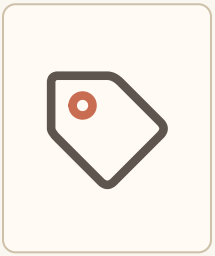
Journal
Diarium



Leaf
a word you know



Cards
a deck



Tag
a label

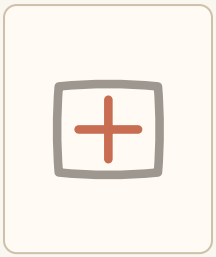
FILES

`docs/handbook/icons/*.svg`: one file each, the line in `currentColor`, the detail in terracotta.

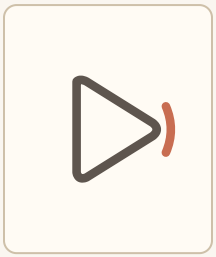
The bookmark is a saved word; the leaf a known one; the stack the library; the journal Diarium. Draw a new learning icon from the desk, not from a screen.

Doing things

The verbs of the app. Where a verb is rare or risky, give it a word as well as a drawing.



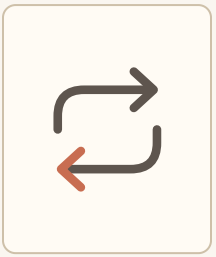
Add
new



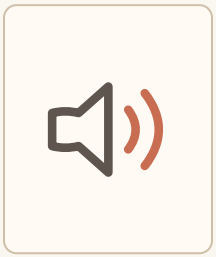
Play
start



Pause
hold



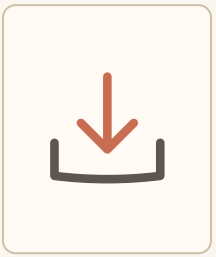
Repeat
again



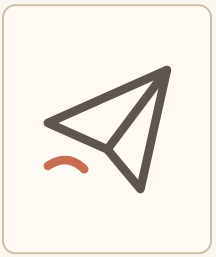
Sound
listen



Microphone
speak



Tray
save to the device



Paper plane
send, share



Waste basket
delete



Undo
take it back



Speech bubble
help, ask

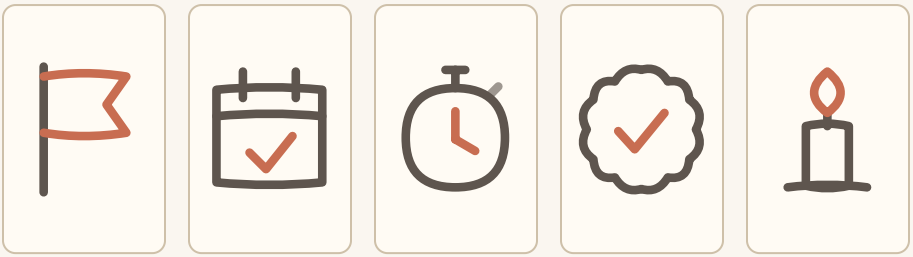


Bell
a reminder

The waste basket carries its terracotta on the lid, not the whole bin: it says “delete” quietly. The paper plane stands for sending or sharing and never for “submit”.

Marks and places

The small things that say how it is going and where you are.



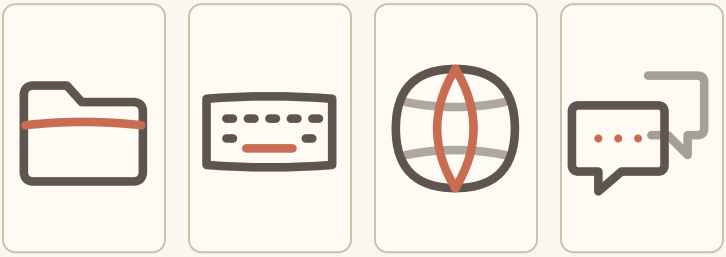
Pennant
a goal

Calendar leaf
a day

Pocket watch
time

Seal
a milestone

Candle
a streak



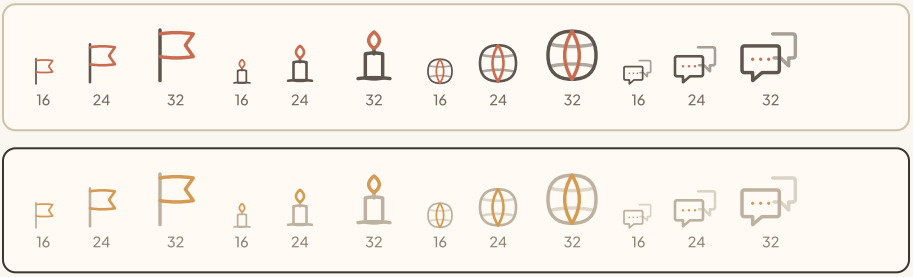
Folder
a group

Keys
typing

Globe
languages

Two bubbles
conversation

A set at small size



Above, 16 to 32 px in Paper; below, the same in Nocturne. If a drawing is unclear at 16, it has too many parts: take one away.

One metaphor, one meaning

ICON	ALWAYS MEANS	NEVER USED FOR
Bookmark	A saved word or place	“Favourite”, “bookmark this page of the app”
Leaf	A word you know; growth	Nature, eco, a leaf-shaped button
Candle	Your streak: days in a row	Birthday, urgency, a warning
Seal	A milestone reached	Verified, approved, a plan badge
Pennant	A goal	Report, flag a problem
Pocket watch	Time spent, or a time chip	Loading
Tag · Folder	A label · a group	Each other
Cards	A deck, review	Payment cards
Speech bubble	Help; ask	Messages from people (Lingua has none)

In the app today

The app’s controls use Lucide icons as lines (15 to 18 px, stroke 1.3), and the rooms their own Lucide icons; these drawings are for the book, the empty pages and the pages the design pass redraws. To bring one into the app, add it as a small component beside the others in `src/design`, drawn with `currentColor` and one accent token, and give it the same name and meaning as here. Nothing in the app uses them yet.

RULE

Never mix the two sets in one toolbar: pick one set per surface.

Where an icon is used, and where it isn't

- **Sidebar and room tabs:** the room's own drawing, with its name always beside it: never an icon alone.
- **Toolbars:** icon with a word for the first three actions; icons alone only for the commonest verbs (play, pause, undo, close), each with a tooltip and an accessible name.
- **Cards and tiles:** one icon, 32 px, at the top left, as the card's kind.
- **Empty pages, onboarding, the end of a book:** a drawing at 96 px or more, made from the same hand.
- **Milestone toasts:** the seal, the candle or the leaf, with the sentence.

DO

- Pair an icon with a word the first time it appears.
- Keep one set per surface.

DON'T

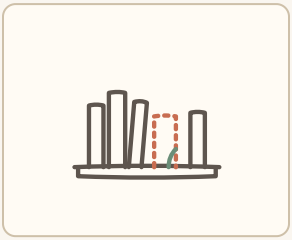
- Invent a second metaphor for something that has one.
- Use an icon to decorate a heading.

Illustration and empty-page art

Where an icon is a word, a drawing is a sentence: it goes where a page has nothing else to say. Same hand as the icons, 96 units wide, the line a little finer (2.4). One small scene, at rest, the terracotta detail on what is missing; no people, faces or mascots, and never on a page with content.



First start
a stack of books, a sprig, a ribbon



Nothing here yet
an empty shelf with a place held



No results
a page and a lens that found nothing



All done for today
a closed book and a cup



No connection
a lamp whose plug has come out



Something tore
a page torn across

DRAWING	WHERE	THE WORDS BESIDE IT
First start	The welcome; an empty Home on the first day	A warm line and one button: “Choose a language”.
Nothing here yet	An empty library, deck, journal, shelf	What this place is for, and the one way to fill it. The dashed book is the one to come.
No results	A search or filter that found nothing	What was searched, and one way out: “Clear the filter”.
All done for today	The end of a review or of Next up	A short, kind sentence, and what tomorrow holds. Never confetti.
No connection	A feature that needs the internet, when offline	What still works without it, which is nearly everything.
Something tore	An error that stops a page	What happened, what it means for them, one thing to do. See “Words on the screen”.

CHAPTER 17

The desktop frame

Lingua is a desktop app. Every screen is drawn for a window 1280 to 1440 pixels wide, with the sidebar always at its left, and has to be just as good on a 1366 × 768 laptop.



1. **Sidebar,** 216 px; it folds away.
2. **The language chip:** the language and its level; the one way to switch.
3. **Room title,** in the language's display face, 42 px.
4. **The one thing,** top right, in the accent fill.
5. **Room tabs,** a segmented control under the title.
6. **The work,** the widest column.
7. **A panel beside it,** 280 to 360 px.

WINDOW	CONTENT COLUMN	WHAT TO DO
1280	about 970 inside the padding	Two columns still fit; three become two.
1366 × 768	about 1055	The laptop: no extra scroll bar; scroll in one place.
1440 and up	1072 (the cap)	The column stops growing; the sides stay empty and quiet.

WHERE

`src/app/shells/desktop`: `DesktopShell.tsx`, `Sidebar.tsx`.

Column `max-w-6xl` (1152 px); sidebar 216 px.

Space comes in fours

Everything is a multiple of 4 px. The few distances that do the work are these, and a new screen should use no others.

TAILWIND

`gap-3` is 12, `gap-5` 20, `p-6` 24.

DISTANCE	WHERE IT IS
4 · 8 · 12	Inside a control, and between small things in a row: icon and label, buttons, stat tiles.
16	Inside a card's small parts; between a label and its field.
20	Between panels, both ways. One number, so a grid of panels looks like one sheet.
24	Inside a panel (its padding), and between a title and what follows.
32 · 40	Between a page's big parts; the page's own side padding is 40.

Columns

- **Two panels side by side** from 1024 px up (`lg:grid-cols-2`), ending on the same line. If one is much taller, move something across, not stretch the short one.
- **Four small tiles in a row** for the numbers that open a page; **three** when each has a chart.
- **A rail** beside the work: filters, the outline, the people, 240 to 320 px wide, sticky under the top, never wider than a third.

The fold

A laptop shows about 560 px below the title: put what a person came to do inside that, and nothing that matters below two screens.

DO

- Let the cap hold: a wide window is calm, not a stretched page.
- Fill height with the work: a list scrolls, a stage stays.

DON'T

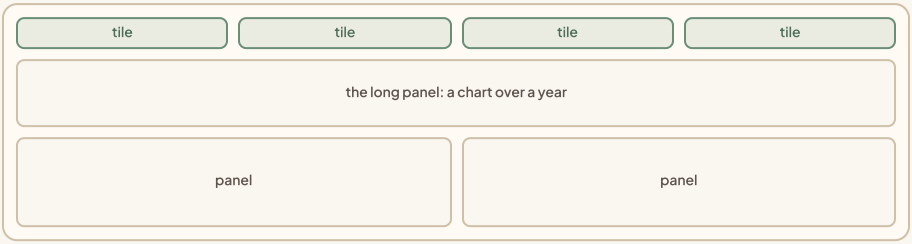
- Stretch a card to the full width of a 1920 screen.
- Make text smaller to make it fit: take something away.

CHAPTER 18

Layout patterns

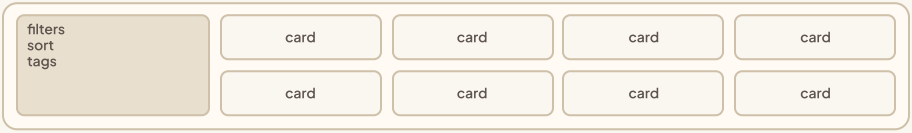
Almost every screen is one of six shapes. Pick the shape first; the details follow.

A The dashboard



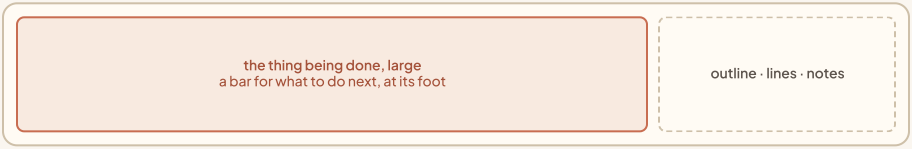
Ratio, Home. Tiles first (the answer in four numbers), then panels, each with a title in capitals, an icon and one action at its right. Used when the page is for looking.

B The list with a rail



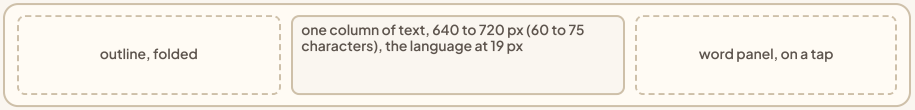
Thesaurus, Bibliotheca, Explora. The search is at the top of the list, not the rail. Used when the page is for choosing.

C The stage with a side panel



Lessons, Ludus, the player, Grammatica. The stage uses the width; the foot bar holds Check and Continue. Used when the page is for doing.

D The reader



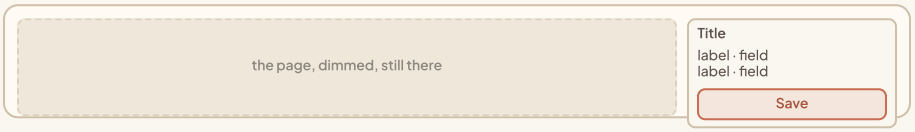
Codex. The text is centred with room either side; everything else is a tap or a key away. Used when the page is for reading.

E The table



Tabula, the inbox, the dictionary list. Rows 48 to 56 px high, the numbers right-aligned in tabular figures, the row's actions at its right and on right-click. Used when the page is a record.

F The drawer



A form that belongs to a place: a goal, log time. It slides in from the right, 440 px wide, 12 px off the edge, its button at the foot. A dialog is only for a question that must be answered.

Choosing

THE PAGE IS FOR…	USE	ROOMS THAT DO
Looking back	A · the dashboard	Ratio, Home
Choosing something	B · list and rail	Thesaurus, Explora
Doing something now	C · the stage	Cursus, Ludus, Visio
Reading	D · the reader	Codex, Diarium
Keeping a record	E · the table	Tabula, Inbox
Adding or changing one thing	F · the drawer	Goals, Log time

CHAPTER 19

Elements, one by one

Every control in Lingua is one of a dozen primitives. Use them as they are: a screen made of them already looks and feels like Lingua.

ELEMENT	SIZE AND SHAPE	WHERE IT GOES, AND WHEN
Button · accent	Terracotta fill, ivory text. 32, 40 or 48 px high (small, medium, large); corners 8, 12, 16.	**The one thing** on a page: top right of the room, or at the foot of a stage. One per screen.
Button · primary	Espresso fill (ivory in Nocturne), a small shadow.	The confirming button of a drawer or dialog, full width at its foot. Also the main button of a card.
Button · secondary	Raised paper, a hairline, darker on hover.	Second to a primary: “Skip”, “Later”, “Choose another”. Never the only button.
Button · ghost	No box; soft ink, a parchment wash on hover; usually an icon first.	Toolbars and the quiet side of a header: Goal, Log time, Image. Cancel is a ghost.
Button · danger	A terracotta outline, no fill.	Delete, disconnect, reset. Never first, never the default focus, and with Undo or a confirmation that names what goes.
Icon button	32, 40 or 44 px square; the icon 15 to 18 px; a tooltip always.	Close, more (···), fold, settings of a card. Four or more in a row means some belong in a menu.
Search	44 px high, corners 12, 15 px text; “/” focuses it.	The top of a list or a room; never in a rail. It searches the page it is on, and says so in its hint.
Popover	Corners 16, the middle shadow.	A small choice that belongs to one control: a sort, a menu, a few options. It closes on a click away or Esc.

ELEMENT	SIZE AND SHAPE	WHERE IT GOES, AND WHEN
Select	44 px, corners 12, a chevron.	A long list of one choice (languages, voices). Fewer than six options: a segmented control.
Segmented	40 px (36 small), a pill that slides inside a rounded track.	Two to five views of one thing: Overview · Tabula, Voice · Text. Never for moving between rooms.
Switch	50 px wide, 200 ms slide, label to its left.	One setting that applies at once. If it needs a Save, it isn't a switch.
Slider	A thin track, a round handle, the value shown beside it.	Speed, volume, size; with the number, and a reset where there's a default.
Card	Corners 24, a hairline, Parchment on Cream; Raised when it holds a choice.	One thing you may open: a book, a deck, a song. A cover or an icon at the top, a title, one line, one number. The whole card is the button.
Panel	Corners 24, 24 px inside, a title in small capitals with an icon.	A group of facts on a dashboard. Its one action sits at the right of its title, as a quiet link with an arrow.
Chip	A pill, 12.5 px bold; Parchment or a wash of its colour.	A tag, a count, a filter. Sage for done or good, terracotta for needs attention, grey for the rest.
Tooltip	Corners 8, 12 px text; 120 ms in, 80 ms out.	Names an icon. Never the only place a fact lives, and never on text that is already readable.

ELEMENT	SIZE AND SHAPE	WHERE IT GOES, AND WHEN
Drawer	440 px (or the screen less 24), 12 px off the edge, corners 24, the top shadow.	A form or details that belong to what's behind it. Its title, a line of help, the fields, one primary button at the foot.
Dialog	Up to 448 px, corners 24, centred, the page dimmed.	A question that must be answered before going on, with two buttons at the most. Never for information.
Toast	Up to 448 px, bottom centre, 24 px up; in and out in 140 ms.	Says what just happened: "Logged 45 minutes of French." With Undo if it can be undone. It goes by itself.
Empty state	Up to 384 px wide, centred, a drawing, corners 16.	A title, one kind line, one button. See "Words on the screen".
Kbd	20 px high, 11 px, a hairline and a tiny shadow.	Beside the thing a key does: "Search /". Alt on Windows, ⌥ on a Mac, as the user chose.

Building a card, from top to bottom

1. **The picture or cover,** a fixed ratio so a row lines up; a drawn icon where there is none.
2. **A small label in capitals:** what it is ("Reading", "Deck"), not what it says.
3. **The title,** in the book's voice, two lines at most, then cut.
4. **One line** of why or how much, and **one number** where it helps: 12 left, 40 min.

DO

- Make a whole card clickable, with a slight lift and a deeper line on hover.
- Keep one primary per card.

DON'T

- Put a button inside a clickable card unless it does something else.
- Stack more than three small details on a card.

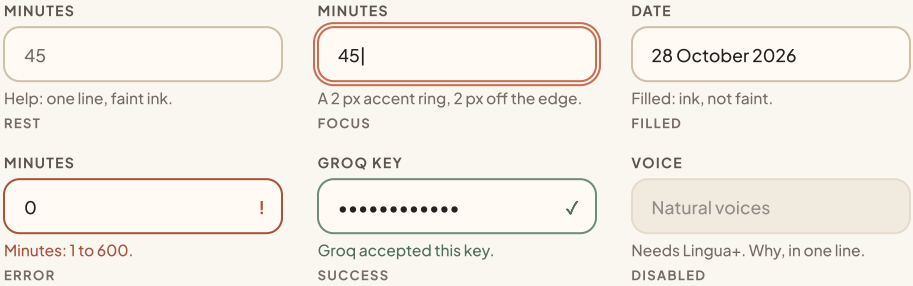
PRESS

Buttons shrink to 97% while pressed, over 150 ms.

Everything is in `src/design/primitives`. If you need a thing that isn't there, add it there for everyone, not in a feature's folder.

Forms and validation PROPOSED

A field is three things: a label above it, a 44 px box (corners 12, 15 px text, a `--line-strong` edge), and one line of help below. That much is built (`h-11 rounded-xl border bg-bg px-3.5` in the drawers). The error and success states below are the design for when a form needs them: today a field either accepts what you type or quietly keeps it within range (Log time stops at 600 minutes), and there is no shared error state or `aria-invalid` yet. When one is added it goes in `src/design/primitives`, for everyone.



PART	THE RULE
Label	Always above the box, 13 px bold capitals in soft ink. Never only a placeholder: it vanishes when you type. Mark the **optional** fields with the word “Optional”, not the required ones with a star.
Help	One line under the box, 12.5 px, faint. It says what goes in, not how the box works. An error **replaces** it, so the height never jumps.
Error	Terracotta text and edge, a “!” at the right, and the words say what is wrong and what is allowed: “That date is in the future. Pick today or earlier.” Colour never carries it alone. Announce it politely and point the box at it with `aria-describedby`.
Success	Only where the answer is slow or doubtful, such as testing an AI key: a sage edge and a tick, and the words. Not on every valid field.
Disabled	Half strength, still readable, with the reason in the help line. If there is no reason to give, the field shouldn’t be there.
When	Check when the field is left, not on every key. On Save, move focus to the first field with an error. Never clear what was typed, and never block typing: a wrong value can be corrected.

CHAPTER 20

Where things go

People look for a thing where they found it last time. Put each kind of thing in its one place, in every room.

THE THING	WHERE	WHY, AND THE RULE
The room's one thing	Top right of the header, in the accent fill	Eyes end the title's line at the right. One per page; if there are two, one is a ghost.
Room tabs	Under the title, at the left	Same place in every room. Five at most; more means a second row of a different kind, or a menu.
Search	The top of the list, full width of it; or the header's right in a room with tabs	It is the first thing tried on a long list. "/" works everywhere.
Filters and sort	A rail at the left (wide), or a row of chips under the search	They change what you see, so they sit beside it, not in a menu.
A list's actions	The right end of each row, ···; the same on right-click	Hover shows them on a desktop; they are never the only way.
Back	Top left, above the title, a small link with an arrow that says where to	"← Thesaurus", not "Back". It goes to the page you came from.
Progress	A thin bar at the top of a stage; a number in the foot bar	It tells where you are, not how you are doing; it never blinks.
Check · Continue	The stage's foot bar, at the right; Skip at the left	The same two spots for every exercise, so the hand learns them.
Settings and help of a thing	A ··· or a gear, or a small Ask button, at the thing's corner	Not in a distant Settings page; Settings holds the app's own.
Messages	A toast at the bottom centre; an inline line where the fault is	The place the eye already is, or the place it must go to fix it.

Keys, menus and the right-click

On a desktop the keyboard is the fast road; the mouse is the road for finding out. Everything that has a key shows it where the thing is.

WHERE

`ShortcutsDialog.tsx` holds the lists; `Kbd` shows a key.

KEY	DOES
Alt 1 … 8 (⌥ on a Mac)	The rooms, in the sidebar's order.
[	Collapse or expand the sidebar.
?	The shortcut sheet, grouped: Navigate, Reading, Studying, Listening, Watching.
/	Search the books or the decks on the page.
Esc	Close a drawer or a dialog.
← →	Previous or next page; in a player, skip back or forward.
Space	Turn the card, then Good; play or pause.
1 … 4 · Z · E	Again, Hard, Good, Easy; undo the answer; edit the card.
K · L · X · D	On a hovered word: known, learning, reset, details. Right-click opens the details too.
Enter	In Ludus, check, then continue.

Menus and right-click

- **A row's menu** is the same list whether you reach it by ⋯ or by right-click, in the same order: the most used first, danger last, after a line.
- **Drag** reorders what has an order (decks, goals, the table's rows), with a grip that shows on hover and the row lifting by its shadow.
- **Double-click** in the reader marks a word as being learned; renaming in place saves with Enter and leaves with Esc.
- **Hover** may add (a ⋯, a preview, a tooltip) but never carries the only way to do a thing.

RULE OF THREE
Title, the work, one action.

Left is for where you are and what you can go to. Right is for what you can do. Under is for what comes next.

The order on a page

Read top to bottom, a page should answer in this order: **where am I** (the room, the title), **what is this** (the one line), **what can I do now** (the work and the one thing), **what else** (beside it), and last, **what is rarely needed** (More, settings, counts). If a screen answers them in a different order, it is probably crowded.

Worked examples: where does it go?

A NEW THING	WHERE, AND WHY
"Export this deck"	In the deck's ··· menu, after the line. It is used rarely and belongs to one deck, so not in the toolbar.
A warning that the AI will use a key	One line under the AI button, in soft ink. Not a dialog: it is information, not a question.
A new filter for Explora	In the rail with the others, in the order of how often it is used. Never above the search.
The progress of a download	A thin bar inside the thing being downloaded; a chip in the header if you leave the page.
"Add a moment" in Ratio	A ghost button at the top right with "Log time": two small, equal, quiet actions; Ratio is for looking.
A tip about a feature	Where the feature is, once, dismissible, in Mundus's voice; never on every visit.

DO

- Put the same action in the same place across rooms.
- Show a key where a thing is, in its tooltip.
- Keep destructive actions at the far end, and quiet.

DON'T

- Hide the only way to do something in a hover or a right-click.
- Move a button because the page below it changed.
- Put "Delete" next to "Save".

CHAPTER 21

How it feels

Feel is a list of small promises kept the same way everywhere: what hovering does, what pressing does, how long things take, what happens when it goes wrong.

STATE	WHAT IT LOOKS LIKE	TIME
Rest	Lines, not shadows. Text in ink; the small lines in soft ink.	—
Hover	The fill goes one step deeper (Parchment on Cream), the line one step darker; a card lifts a little. The cursor is a pointer only on things that act.	150 ms ease-out
Press	A control shrinks to 97% (a tab-bar item to 94%) and the fill deepens.	150 ms
Focus	A 2 px terracotta ring with a 2 px gap, on every control, for the keyboard only. Never removed.	at once
Selected	A wash of the accent or sage behind it and the text in its dark; the segmented pill slides to it. Never colour alone: also a tick, a weight or a line.	snappy spring
Disabled	45% opacity and no pointer events; its reason in a tooltip if it isn't obvious.	—
Loading	The shape of the thing, softly pulsing (a skeleton), for anything over about 200 ms; a spinner only inside a button that was pressed.	after 200 ms
Arriving	A page rises 8 px and fades in; a drawer slides in; a toast rises 8 px.	soft · sheet springs
Done	A small change where it happened (a tick, a number counting up), and a toast only if it is somewhere else.	under 1 s
Wrong	A line next to the thing, in terracotta text with a plain sentence. The page never shakes, and never says "Error".	—

The springs are in `src/design/motion.ts`; only transform and opacity move. Reduced motion removes the movement entirely and keeps the fades.

SOUND

Off unless the learner asked for it; a click for a right answer is the most it does.

Speed is part of the feel

WHAT	SHOULD TAKE
A tap or a click answering	under 100 ms, always visibly
A page showing something	its frame at once; its content under 300 ms from the device's own data
A word card after a click on a word	under 100 ms; the translation may follow
Opening the reader on a book	under 500 ms to the last page
Anything that waits on the internet	It says so, shows progress, and can be left

Making a wait kind

- **Show the frame first,** then fill it. A page that appears empty and then jumps feels broken; one that appears shaped and then fills feels fast.
- **Say what is happening** in the helper's voice: "Finding the sentences…", never "Loading".
- **Let people leave.** A long job runs on, and a small chip says it's still going.
- **Keep their place.** Going back returns to the scroll position, the tab and the filter they had.

The small moments

MOMENT	HOW IT SHOULD FEEL
A right answer	A quiet sage tick that draws itself in 200 ms, the foot bar turning sage; a soft click if sound is on. No confetti.
A wrong answer	Terracotta, the right answer shown at once beside it, and Continue waiting. No buzz, no shake, no lost hearts.
Turning a card	A flip of 200 ms with the shadow following; Space does it.
Marking a word known	Its underline fades from terracotta to nothing, in the reader, where the eye is. No toast.
Saving a word	A small “Saved” in place of the button, and a number going up by one in the deck chip.
Finishing a lesson	The stage settles and a quiet summary rises: what you did, how long, what is next. One line of warmth in the helper’s voice.
A milestone	One toast with a small drawing: “100 words known.” It never pops up over reading or listening.
A missed day	Nothing. Lingua doesn’t scold, and a streak never breaks loudly.

When it goes wrong

SAY

- What happened, in a sentence: “The recording didn’t start.”
- What it means for them: “Your words are safe.”
- One thing to do: “Try again”.

NEVER SAY

- A code or a stack, “Something went wrong”, or “Oops”.
- Anything that blames them.
- Nothing, if their work might be lost.

Pointer, keyboard, screen reader

Each is a first-class way in: every control reachable with the keyboard in reading order, every icon named, colour never the only signal, text that can grow without breaking a page.

The check in Part Five has the list; do it before you call a page done.

CHAPTER 22

Dark, and the papers

Nocturne isn't Paper turned upside down. It is the same room with the lamp low: the same layout, the same hierarchy, warmer and softer, never pure black or pure white.

ROLE	PAPER (DAY)	NOCTURNE (NIGHT)	USED FOR
Page	Cream #faf6f0	Charcoal #1f1c1a	The ground of every page.
Surface	Parchment #efe7d9	Mocha #2b2724	Panels, the sidebar, tiles.
Raised	#fffbf4	#35302c	Fields, cards holding a choice, menus.
Ink	Espresso #262220	Ivory #ebe0cc	Text; the primary button's fill, swapped.
Accent	Terracotta #c86d51	Amber #d49b54	The one thing, learning, selected.
Sage	#6b8e78	#93b39e	Known, done, good.

Contrast, measured

PAIR	RATIO	PAIR	RATIO
Ink on Cream	14.6 : 1	Ink on Charcoal	13.0 : 1
Soft ink on Cream	6.8 : 1	Soft ink on Charcoal	8.0 : 1
Faint ink on Cream	4.7 : 1	Faint ink on Charcoal	4.8 : 1
Faint ink on Parchment	4.2 : 1	Faint ink on Mocha	4.2 : 1
Terracotta text on Cream	5.0 : 1	Amber text on Charcoal	7.8 : 1
Sage text on Cream	5.2 : 1	Sage on Charcoal	7.4 : 1

TOKENS

`src/design/tokens.css`: `--bg` `--surface` `--ink` `--accent-text`.

Body text needs 4.5 : 1. **Faint ink falls just under it on Parchment and Mocha:** use it there only for large or purely decorative text, and soft ink for anything that must be read. The bright terracotta line (3.4 : 1) is for lines and fills, never for small text; text uses the darker `accent-text`.

Making a screen that works in both

- **Never write a colour.** Use the token: `bg-surface`, `text-ink-soft`, `border-line`. A hex in a feature's code is a bug in one of the two themes.
- **Shadows get heavier at night,** not lighter: on dark ground a light shadow is invisible, so lift with a lighter surface and a hairline too.
- **Pictures need care.** Covers and photographs can glare on Charcoal: a quiet scrim or a slight dim at night, never a filter that changes their colours.
- **Drawings use the line colour,** so they flip by themselves; a drawing with its own colours must be checked on both.
- **Accent flips from warm terracotta to amber,** on purpose: the same feeling at a different light. Never use the day accent at night.
- **Selected, learning and error** must read without colour in either theme: a weight, an underline, a tick.

The reading papers

WHERE

`src/design/papers.ts`.

The reader chooses a page colour for Codex separately for day and for night: seven light papers and seven dark ones. It changes the page, the panels and the ink together and flips with the theme.

DAY	PAGE	INK	RATIO	NIGHT	PAGE	INK	RATIO
Cream	#faf6f0	#262220	14.6	Charcoal	#1f1c1a	#ebe0cc	13.0
Ivory	#fdfbf6	#23211f	15.5	Espresso	#241b16	#eedcc1	12.6
Parchment	#f5ecdb	#2d2520	12.8	Sepia night	#2b2219	#efdcb9	11.6
Sepia	#f1e5cc	#3b2d21	10.6	Ink	#1a1c21	#e8e2d6	13.2
Sage	#eff1e7	#252a23	12.8	Forest	#1b211d	#e6e4d3	12.8
Blush	#faeee8	#2d2321	13.4	Plum	#221c22	#eee1dc	13.1
Stone	#f2f0eb	#242320	13.8	Midnight	#100f0e	#e4d7c2	13.5

Ratios are each paper's ink on its own page, worked out from `papers.ts`: all fourteen clear 4.5 : 1 for body text, the lowest being Sepia at 10.6. Faint and accent text on a paper must still be checked by eye, because only the ink is listed here.

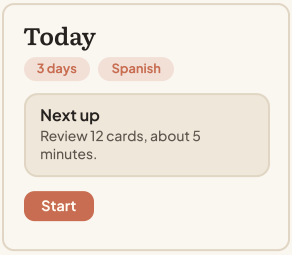
A screen must hold on all fourteen. When designing, switch through the papers, not only the two defaults: contrast and the accent must hold on each, and nothing may rely on Cream's warmth to look right.

Switching

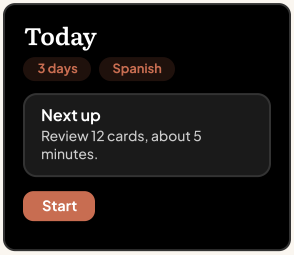
- Light, dark or the device's own, in Settings › Look and feel; the choice crossfades the whole window in one view transition, not piece by piece.
- The first paint follows the device's setting before the app loads, so there is no flash of the wrong colour.
- The browser's top bar and the installed app's title bar take the page colour.

A screen in Nocturne: wrong, then right

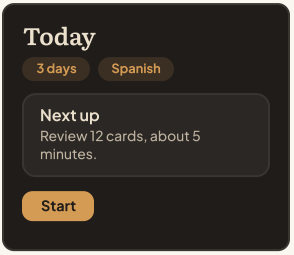
The same card in three lights. The middle one is what happens when Paper is simply inverted; the third is Nocturne as designed.



Paper. Cream, Espresso, Terracotta.



Inverted: wrong. Pure black and white glare; the card vanishes; the accent is dull.



Nocturne. Charcoal, soft Ivory, Amber; the card is lifted by Mocha and a hairline.

WHAT CHANGED	PAPER → NOCTURNE	WHY
Ground	Cream #faf6f0 → Charcoal #1f1c1a, never #000	Pure black makes every edge shimmer; a warm dark rests.
Text	Espresso → Ivory #ebe0cc, never #fff	White on near-black is harsh over a long read.
Cards	Parchment → Mocha #2b2724, plus a hairline #3b3532	A shadow is invisible on dark, so lift with a lighter surface.
Accent	Terracotta #c86d51 → Amber #d49b54; button text flips to Charcoal	Terracotta loses contrast at night; amber keeps the feeling.
Sage	#6b8e78 → #93b39e	Lighter so that “done” still reads.

Checking a screen at night, in order

1. Is there any pure black or white, or a hex in the code? There must be none.
2. Does every card still read as a card without its shadow?
3. Do the accent button and its label keep 4.5 : 1? (Amber on Charcoal is 7.8 : 1.)
4. Are pictures dimmed a little, and their colours untouched?
5. Squint, or turn the screen to grey: is the hierarchy the same as by day?

CHAPTER 23

Charts and numbers

A number is evidence, not a grade. Show it plainly, say what it counts, and draw a chart only when a shape says more than the number.

The stat tile



The number big, in the book's voice; a caption in small capitals under it saying what it counts; one line of context in soft ink. Four of them open Ratio.

Which chart

THE QUESTION	CHART	RULES
How is it changing?	Line	Time along the bottom, from the start of the data. No more than three lines, each named at its end. Sage for known, terracotta for learning. A short gap is carried across, a long one left empty.
How much in each period?	Bars	One bar per day, week or month; the current one outlined. Stack by kind (read, listen, review…) only when the parts matter. A baseline at zero, always.
How regular?	Heat map	A year as weeks and weekdays, five shades of sage; today outlined. The scale in a small legend.
How far to a goal?	Ring or slim bar	One value of one whole, with the figure in the middle or beside it. Never two rings side by side that mean different things.
What is it made of?	A row of bars with labels	Sorted by size, labels at the left, values at the right. Not a pie.

HEIGHTS

Lines 110 to 130 px; long bars 110; rings 58.

The charts are in `src/design/charts` (`LineChart`, `BarChart`, `Heatmap`, `Ring`), each with a hover that names the value and date.

Colour in a chart

- **Meaning first.** Sage is what you have (known, done); terracotta is what is moving (learning, today). The same colours mean the same things on every page.
- **Rooms get tints, not hues:** reading is sage, listening a paler sage, watching and cards warmer, writing nearly paper. Seven tints are enough, and any two must be tellable apart in grey.
- **Always label directly** where a line ends or a bar sits; a legend is for stacks only, and sits above the chart.

Numbers in sentences and tables

DO	BECAUSE
2,400 words; 1.5 h; 12 min	Thousands separated, time in the unit people think in, never 5400 s.
Tabular figures in tables	The digits line up and the eye compares.
“Words / min”, not “WPM”	The caption says what it counts.
~A2 and “estimated”	An estimate is labelled as one, with how it was made.
“Show once you have read a little more”	A figure from too little data would be a guess dressed as a fact.

Empty, thin and sudden

DO

- Say kindly why it's empty and what fills it: "Check two entries to see your grade trend".
- Keep the chart's frame so the page doesn't jump when data arrives.
- Show a single point as a point, not a line.

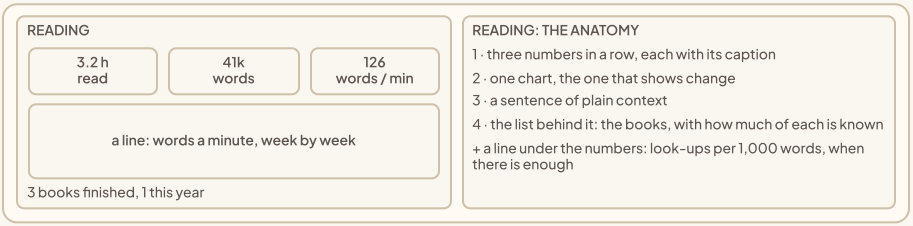
DON'T

- Draw a flat line for no data.
- Compare to other people, or to a number the learner didn't choose.
- Colour a day red for a missed one.

Honest numbers Lingua keeps

- A **known word** is one the learner said they know; a saved word is saved, and is never called known.
- **Time** counts active time with the window visible and a recent action, so a book left open isn't an hour read.
- **Look-ups per 1,000 words** appear after 500 words are read, from sessions that counted them.
- **Retention after a day** isn't recorded, so it isn't shown; a number Lingua can't back isn't made up.

A worked example: Ratio's Reading panel



Every panel on Ratio is built the same way: caption and numbers, one chart, a line of context, and the list behind it. That regularity is why a dashboard of eight panels still reads as one page.

Anatomy of a chart

Every chart in Lingua (`src/design/charts`: line, bar, calendar heat map, ring) has the same parts, in the same places. A chart missing one of them is unfinished.



#	PART	HOW IT IS MADE
1	*Title and line*	What is counted, with its unit, in 12 px bold; and one faint line for what it leaves out. “Minutes a day”, not “Activity”.
2	*Scale*	Two or three ticks, in faint ink, as plain numbers. A bar chart starts at zero. A line may start higher, but then says so.
3	*The last value*	A dot and the number beside it: where you are now matters more than the whole curve.
4	*Labels on the lines*	Name the line where it runs, in its own colour, instead of a legend to look up. A legend only when there are labels that cannot fit.
5	*Series*	Three at most: terracotta, sage, soft ink. A line is 1.7 px, round-capped; no points except the last.
6–8	*Grid, baseline, dates*	Hairlines, a slightly stronger baseline, and 3 or 4 dates, “Today” at the right end. No box around the chart, no 3D, no shadows.

Reading one: what is counted, over what time, and is the last point partial? **Hover** shows the exact number and date in a tooltip; it is a bonus, never the only place a number lives. **For a screen reader** each chart is one image with a label that gives the latest value per series (“Reading: 42 min now”). The numbers behind it as a table, on request, would close the gap: PROPOSED

WHERE

`LineChart`, `BarChart`, `Heatmap`, `Ring` in `src/design/charts`.

CHAPTER 24

Words on the screen

Most of the design is words: a button's name, an empty page, a warning. They are set in the helper's voice: plain, warm, short, and on the learner's side.

WHERE	INSTEAD OF	WRITE
A button	Submit · OK · Yes · Proceed	A verb and what it acts on: **Log 45 minutes**, **Save the word**, **Start the lesson**.
A cancel	Cancel	**Not now**, **Keep it**, **Go back**: say what stays true.
A delete	Are you sure?	**Delete "Cher journal"?** Say what goes; give Undo, not a second question, where it can.
A title	Settings Management Console	The name a person would use: **Settings**, **Tabula**, **Your words**.
The line under a title	Here you can view statistics about…	One sentence in the learner's terms: **An honest look back at what you've done.**
A toast	Operation completed successfully	**Logged 45 minutes of French.** Past tense, the thing, then stop.
An error	Error 500 · Invalid input	**That didn't save. Your words are safe. Try again.**
A hint in a field	Enter text	An example in the language: **e.g. Dutch, Arabic**.
A price or plan	Premium · Unlimited · Best value	The price, what it includes and when it renews, together: **€24.99 once, every update included.**

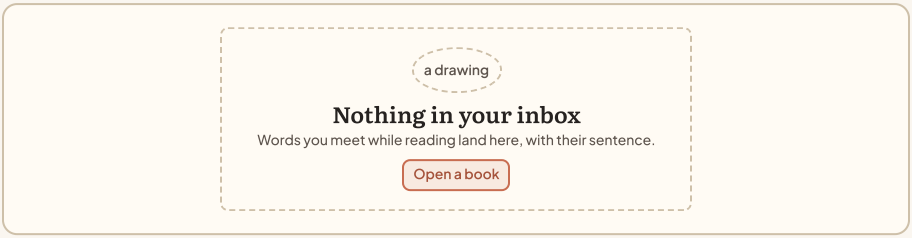
CASE

Sentence case everywhere. Capitals only in the small labels, set by the design, not typed.

The voice has its own chapter in Part One; this is its use on the smallest surfaces. A button that has an icon still has a name, and the name is what a screen reader says.

The empty page

An empty page is the first thing a new learner sees of a room, so it is written with care. It has exactly three parts, and the third is a button.



1. **A title that says what is true,** not what's wrong: "Nothing here yet" is fine; "No data found" is not.
2. **One line that says what will fill it** and how, in the helper's voice.
3. **One button that does the first thing,** in the accent fill.

Six empty pages, written

PAGE	TITLE	THE LINE	THE BUTTON
Inbox	Nothing waiting	Words you meet while reading land here with their sentence.	Open a book
Decks	No decks yet	A deck is a list of words to remember; start with a ready-made one.	Browse decks
Diarium	Today…	A few sentences in the language are enough.	Start writing
Visio	Nothing to watch yet	Add a film, a video or a link, and its subtitles go in two lines.	Add a video
Goals (Tabula)	No goal yet	A level by a date, or hours a week. You can change it any time.	Set a goal
A search	Nothing for "xyz"	Try fewer letters, or search in the language you're learning.	Clear the search

Rules for all small words

DO

- Write the way a kind teacher would say it aloud.
- Use the learner's word: "word", not "lexeme"; "deck", not "collection".
- Put the most important word first.
- Keep a button to one to three words.
- Name a thing the same everywhere: it's always "Tabula".

DON'T

- Use "please", "sorry" and "oops" as padding.
- Praise for nothing, or shout: no "Amazing!" or exclamation marks.
- Make a joke where something went wrong.
- Use "we" for the app; Lingua says "I" rarely and "you" often.
- Leave a word for translation: all text is ready for the language of the app.

The app's language, and the learner's

The app speaks the learner's own language (the one glosses and help come in) and shows the language being learned wherever the learning is: in the reader, the cards, the lesson. Do not mix them in one line without a visual reason: the language being learned is set in the book's voice and larger, so you can always tell it from the helper's.

Texts may grow by a third in another language: a button must never be sized to its English.

CHAPTER 25

Accessibility

A language learner reads small text, listens closely and often studies tired. Making Lingua usable without a mouse, without sight, without hearing, or without seeing colour makes it better for all of them.

AREA	WHAT LINGUA DOES TODAY	STILL TO DO
Focus	Every control shows a 2 px accent ring, 2 px off its edge, for keyboard focus only (`:focus-visible`).	—
Names	Icon buttons need a `label` and carry it as their accessible name. Switches are `role="switch"`; segmented controls are radio groups; the first start says “Step 3 of 7” and marks the current step.	Name the new icons as the icon chapter says.
Overlays	Drawers are modal dialogs, the confirm box an alert dialog: Tab stays inside, Esc closes the topmost, focus goes in and comes back. A toast is announced (`status`; an error as `alert`).	—
Motion	Settings › Motion: Device, Calm or Lively (System, Reduced or Full in the full view). Device follows the system’s “reduce motion”; the choice is on the page as `data-motion`.	—
Keyboard	Alt 1–8 for the rooms, **?** for the shortcut sheet, **/** to search, **[** for the sidebar, letters on words; every drawer by Esc.	A skip-to-content link was not found in the code. PROPOSED
Text size	The interface follows the browser’s zoom. Reading text has its own size, 15 to 30 px, and line spacing (Settings › Reading). 12.5 px is the smallest interface text.	The layout at 200% zoom has not been measured.
Colour	State is never colour alone, as a rule. Body text clears 4.5 : 1 except where measured below.	Two findings, over the page.

Two things we measured

PAIR	NORMAL	PROTAN	DEUTAN	WHAT IT MEANS
Terracotta #c86d51 · Sage #6b8e78 (Paper)	1.0 : 1	1.2 : 1	1.1 : 1	The same lightness. Only the hue differs.
Amber #d49b54 · Sage #93b39e (Nocturne)	1.1 : 1	1.2 : 1	1.0 : 1	The same at night.
Accent text #a5533b · Sage text #4f6f5b	1.0 : 1	1.2 : 1	1.2 : 1	Text in either looks equally dark.
Faint ink on Parchment · on Mocha	4.2 : 1	—		Just under 4.5 : 1.

The simulations are worked out with a standard colour-vision matrix and show the direction, not a verdict: check with a tool as well. **The consequence is firm:** terracotta ("needs attention") and sage ("done") can look the same to a colour-blind learner, so each must always come with a second signal. That is the rule in "Dark, and the papers", made concrete:

- A tick or a word with sage: "Known ✓".
- A mark or a word with terracotta: "Due", "!".
- In a chart, label the lines and keep the two apart in position, not only in colour.
- Learning, known and unseen words in the reader differ by underline and weight as well as tint.
- Faint ink only for large or decorative text on Parchment and Mocha; soft ink for what must be read.

Focus order, and the names of things

THING	ORDER OR NAME
Tab order	The reading order: sidebar, then the header (back link, title, tabs, the one thing), then the content left to right and top to bottom, then a side rail, then the foot bar. Never `tabindex` above 0.
An open drawer	Focus goes to the first field marked `data-autofocus`, or to the drawer; Tab cycles inside; Esc closes and returns focus to what opened it.
An icon on its own	Its meaning from the icon chapter: "Bookmark: a saved word". With a word beside it the icon is hidden from screen readers.
A drawing	Decorative: hidden from screen readers. The empty-page words say what the drawing says.
A flag	Never alone: the language's name always sits beside it.
A chart or a ring	An image with a label: "Reading: 42 min now", "Goal: 12 of 20 minutes".
A chip	The word is in the chip; colour only adds.
The learning language	Text in the language being learned should carry its `lang`, so a screen reader speaks it right. PROPOSED to check where it is set.

Naming recipe

SAY WHAT IT DOES	NOT	THE RULE
"Close"	"X", "cross"	The name is the action, not the picture.
"Mark as known"	"Green tick"	Never the colour or the shape.
"Play, 12 s of 3 min"	"Button"	Include the state that changes: playing, off, 3 of 10.
"Reading: 42 min now"	"Chart"	A chart is named by what it shows, and its latest number.
"Spanish" beside a flag	"Flag of Spain"	The language, not the flag. The flag is hidden.

- **Short:** four words if it can be, and no "button" or "icon" (the role is announced).
- **The same everywhere:** one meaning, one name, in every room. The icon chapter's table gives the names.
- **In the learner's language,** not the language being learned, except for the words being learned.

Checks to run on every screen

- ☐ **Keyboard only.** Reach everything, see the focus, open and close every drawer.
- ☐ **200% browser zoom** at 1280 px: nothing cut off, nothing overlapping, no sideways scroll.
- ☐ **Reduce motion** on: no page slides, no springs that matter; the meaning still arrives.
- ☐ **Grey.** Turn the screen to greyscale: is every state still readable?
- ☐ **Contrast.** Body text 4.5 : 1, large text 3 : 1, in Paper, Nocturne and the papers it can sit on.
- ☐ **A screen reader.** NVDA on Windows (free), VoiceOver on a Mac (Cmd F5), TalkBack on Android. Listen to one screen end to end.
- ☐ **Names.** Every icon button, chart and switch is announced with what it does.
- ☐ **Errors.** They are announced, say what is wrong, and put focus where to fix it.
- ☐ **Sound off, voice off, AI off.** The screen still works and still makes sense.
- ☐ **Targets.** 32 px at the least with a mouse, 40 to 44 under a finger.

Reduced motion, in detail

CHOICE	WHAT HAPPENS
System	Follows the device: if it asks for less motion, Lingua moves less too. The default. (“Device” in the simple view.)
Reduced	Calm and still, whatever the device says: things fade in place instead of sliding or springing.
Full	Rich and liquid: pages glide in, lists reorder smoothly, everything settles softly; smoothest on 120 Hz. Never required to understand anything. (“Lively” in the simple view; Reduced is “Calm”.)

TEST

Turn it on and click through the new screen.

Rule for new motion: read `motionIsReduced()` (`src/design/motionPref.ts`) or use the springs in `motion.ts`, and make the reduced version a short fade or nothing. Meaning must never ride on movement alone.

CHAPTER 26

Sound and haptics

Lingua is mostly quiet. Sound has two jobs: it is the content (a voice, a song, a station), and, in a few places, it says how an answer went. Only the second is a design matter, and it should be small.

What exists. Six cues, made in the browser with Web Audio (no sound files), played in the lesson sets of Ludus and the grammar drills. The learner can switch sounds off in Settings (“Sounds in lessons”) and sounds and haptics in the lesson setup sheet; both start on. Vibration works where the browser lets it (Android Chrome; not iOS Safari, not a desktop), so on the desktop only the sounds are heard.

WHERE

`src/features/lesson/sfx.ts`, used by the lesson kit.

CUE	SHAPE	MEANS	VIBRATION (MS)
right	E5 then B5 (660, 990 Hz), sine, 0.3 s: a rising fifth	A right answer	12
wrong	A♯3 then G3 (233, 196 Hz), triangle: two low falling notes	A miss, gently	30 · 40 · 30
combo	E5, G♯5, B5, E6: a four-note run up, 0.34 s	A streak in a row	10 · 30 · 10 · 30 · 18
finish	C5, E5, G5, C6: an upward arpeggio, last note 0.5 s	The lesson is complete	20 · 60 · 40
tick	1760 Hz, 50 ms, very quiet	A counting step	none
tap	880 Hz, 60 ms, very quiet	A press that deserves a click	none

The loudest cue is at a gain of 0.12, below speech and music; ticks and taps at 0.05. Notes begin in 12 ms and fall off exponentially, so nothing clicks.

Where it must be silent RULES

The cues are built; these are the rules for any new one. When in doubt, no sound.

WHERE	WHY IT IS SILENT
Opening a page, switching a room, loading	Nothing has been judged. A cue means “you answered”.
While audio plays	A song, a podcast, a station, a voice: the sound is the content. A chime on top of a sentence spoils it. Pause or skip the cue.
While the microphone listens	Speaking exercises: the cue would be recorded, and it would break the learner’s concentration. Play it after the recording stops.
Reading and writing	The reader, Diarium and typing: no cue for finishing a page or a word. Silence is the room’s whole point.
Ordinary buttons and menus	Clicks are not events. Only a judged answer, a streak or a finish earns a sound.
Errors in forms, toasts, settings	A warning tone is an alarm; use words.
When sounds are off	Silent everywhere, with no hidden exceptions, and the vibration follows its own switch.

Should it make a sound?

MOMENT	SOUND	WHY
A card answered right, or wrong	right · wrong	A judged answer.
Five right in a row	combo	A streak is the reward.
A timer counting down in a game	tick	It counts; keep it very quiet.
A lesson finished	finish	The one place for a small flourish.
A word saved from the reader	none	A toast says it; the reader is quiet.
A drawer opened, a tab switched, a page loaded	none	Not an event.
A goal reached for the day	none	A line of warmth in words; sound would arrive unasked.
A typing mistake in a typing drill	none	It would sound on every key.

The language of the cues

- **Up means well, down means miss,** and more notes mean more: one rising pair, a run, an arpeggio. A learner should be able to tell right from wrong with eyes closed.
- **A miss is soft,** low and short, never a buzzer. Nothing in Lingua should make a learner flinch.
- **The reader's buzz:** on Android, holding a word for 430 ms gives an 8 ms tap as the drawer opens. That is the only vibration outside a lesson.

Making a new cue

1. **Ask whether it is needed.** If the screen already says it, a sound is a second voice.
2. **Build it from the existing family:** sine or triangle notes, a few of them, a final note under 0.5 s.
3. **Keep it under the music:** peak gain 0.12 at the most.
4. **Give it a vibration only if it matters**, shorter than 100 ms in all.
5. **Write it into the table here,** with what it means, and test it with the sounds off.

DO

- Make the off switch one click, in the place the sounds are heard.
- Play cues after the visible result, not before.

DON'T

- Play a sound that starts by itself, ever.
- Make a haptic the only signal: most devices have none.

Web Audio needs a click or a tap before it will play, which a lesson always has. The speaking voices (recordings, the device's voice, Piper) are separate: they are content, and they are covered in Part Two.

CHAPTER 27

Designing a new screen

A recipe that works every time. Follow it in order; each step takes minutes, and skipping one is where most mistakes begin.

1. **Write the sentence.** “This page is for ___.” One verb; if you need “and”, it is two pages, or a tab.
2. **Pick the shape** from the six patterns: dashboard, list with a rail, stage, reader, table, drawer.
3. **Choose the one thing,** the single action the page leads to, and where it goes. Everything else is a ghost or a link.
4. **List what to show first and what to keep quiet,** as in the room cards: what is on the page at once, and what waits behind a tab, a hover or More.
5. **Draw it in boxes** at 1366 × 768 with real content and the longest word; then at 1440 and 1280.
6. **Build with the primitives** and the tokens; add nothing new unless you have asked for it in the primitives.
7. **Write the words** last but before it is called done: the title, the line, the buttons, the empty page, the errors.
8. **Put it through every state** (next chapter in Part Five): empty, loading, partial, full, error, offline, slow.
9. **Check it** in Paper and Nocturne, with the keyboard only, with the text larger, and with the voice and AI off.
10. **Show it to one learner** and watch without helping; count the pauses.

Questions that settle most arguments

IF YOU'RE NOT SURE WHETHER TO…	ASK
add a button	Does anyone need it in the first minute? If not, it goes in ···.
add a setting	Can it be decided for them? A setting is a failure to decide, unless people truly differ.
show a number	Would the learner act differently because of it?
add motion	Does it say where something came from? If not, it is decoration.
use a new colour	Is there already a token that means this? There is.
add a dialog	Is it a question that must be answered? Otherwise it is a toast or a line.

The brief to fill in before drawing

Copy this onto the top of the step's plan. If a line can't be filled in, the screen isn't ready to draw.

This page is for _____

Room, and shape _____

Who, and when _____

The one thing _____

_____

Shown first _____

_____

Kept quiet _____

The words _____

States and plans _____

Done when _____

WHERE

Put them in the step's plan in `docs/`.

Handing a screen over

A design is finished when someone else can build it without asking you. Hand over these, in this order.

1. **The sentence and the shape:** "For choosing a deck; list with a rail."
2. **The boxes,** at 1366 × 768, annotated with the primitives and the tokens.
3. **What is shown first and what is kept quiet,** written as in a room card.
4. **The words,** every one, including empty and error.
5. **The states:** empty, loading, partial, full, error, offline.
6. **What is checked:** the five things to try by hand before calling it done.

The same brief, filled in: Log study time

This page is for	Adding time you spent learning away from Lingua.
Room, and shape	Ratio, in a drawer (440 px), opened from Overview and from Tabula.
Who, and when	Someone who just finished a class or a podcast, in under twenty seconds.
The one thing	"Save" at the foot of the drawer, primary, full width.
Shown first	Language, what (class, tutor, other app), and how long: chips for 15 to 90 minutes and a number field. The date is today.
Kept quiet	The date, and a note marked "Optional", below the fold of the drawer.
The words	Title "Log study time", help "Classes, tutors, conversations, other apps: it all counts." Minutes 1 to 600. Toast: "Logged 45 minutes of French." with Undo.
States and plans	Minutes are held to 1–600, with no error text yet; a toast on save. Free: all of it.
Done when	Paper and Nocturne at 1280, 1366 and 1440; keyboard only from open to save; the time shows in Ratio.

The twenty mistakes we keep meeting

#	THE MISTAKE	THE FIX
1	Two accent buttons on a page.	One accent; the other is a ghost or a link.
2	A card in a card in a card.	A card holds text and one picture; group by space first.
3	A shadow on everything.	Lines for rest; a shadow only for what floats.
4	Text set smaller to make it fit.	Remove something; 12.5 px is the floor.
5	A hex code or a grey in a feature's code.	Use the tokens.
6	Colour as the only signal of state.	Add a tick, a weight or an underline.
7	The only way to do it is a hover or a right-click.	Make it visible, then add the shortcut.
8	A button that moves when the page changes.	Reserve its place; keep the frame.
9	"Loading…" on a blank page.	Show the shape, then fill it.
10	An error that says "Error".	What happened, what it means for them, one thing to do.
11	Delete next to Save.	Danger at the far end, quiet, with Undo.
12	A stretched card on a wide screen.	Let the 1152 px column hold.
13	A scroll bar inside a scroll bar.	One scrolling area per column.
14	An icon without a name.	A tooltip and an accessible name, always.
15	Praise on every step.	Quiet confirmation; a word of warmth at the end only.
16	The one thing at the bottom of a long page.	Top right, or sticky in the foot bar.
17	A page tested only with short, English text.	Test the longest word and a right-to-left language.
18	Three tab rows.	One row; the rest in a rail or a menu.
19	New patterns invented for one screen.	Use one of six, or change the pattern for all.
20	Called done without the check.	Part Five's check, every time.

Add your own to this list when you meet a twenty-first. It is the cheapest design tool Lingua has.

CHAPTER 28

Tablet: iPad and Android

GUIDANCE ONLY Lingua is built and checked for the desktop. A tablet is the next-nearest thing: the same rooms on a bigger screen you touch, and a very good place to read, listen and review.

RULE

Desktop first. Phone and tablet are guides for later, not a reason to change a desktop screen.

Where it stands. The app already has a second shell for narrow windows (below 768 px, see the phone chapter), and the controls already grow under a finger. What hasn't had its own design pass is the tablet, so this chapter is the brief for it: how it should look, feel and be laid out when it gets one.

Which size is it?

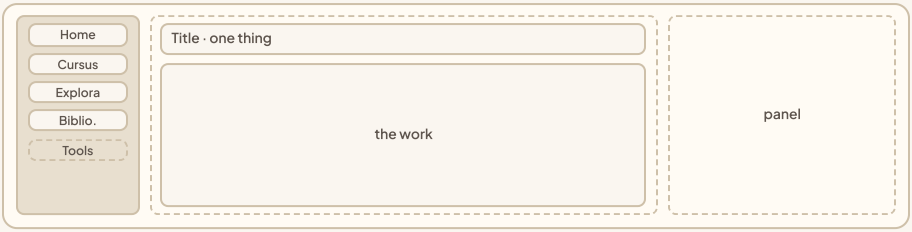
DEVICE (CSS PIXELS)	PORTRAIT × LANDSCAPE	LAYOUT IT GETS TODAY
iPad mini	744 × 1133 · 1133 × 744	Compact in portrait; the sidebar in landscape
iPad, iPad Air 11″	820 × 1180 · 1180 × 820	Sidebar both ways, narrow in portrait
iPad Pro 11″	834 × 1194 · 1194 × 834	As above
iPad Pro 13″	1024 × 1366 · 1366 × 1024	As the desktop at 1024 and 1366
Android tablets	about 800 × 1280 · 1280 × 800	As the iPad; check on a real one

WHERE

`src/design/hooks/useLayout.ts`.

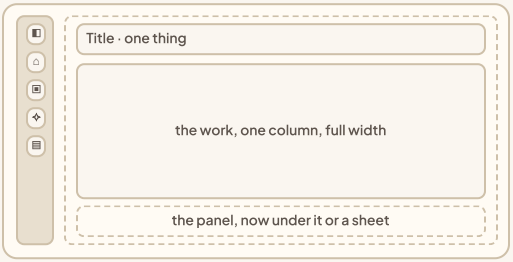
Today the shell switches at 768 px: from there up it shows the desktop sidebar, whatever the pointer; below it, the phone's top bar and tab bar. The pointer is read separately (`fine` or `coarse`), so a tablet gets the desktop layout with bigger targets. Stage Manager and Split View can narrow the window at any time, and it should simply change shell.

Landscape: the desktop, touched



1180 × 820. It is the desktop layout at the width of a small laptop: sidebar, work, panel. The changes are in the hands, not the grid: every target 44 px or more, nothing that needs a hover, and a bottom margin for the home indicator.

Portrait: the sidebar steps back



820 × 1180. The sidebar folds to a rail of icons (about 64 px) with names on press, so the work keeps over 750 px. A side panel drops under the work, or into a sheet; two columns become one, except when each is narrow (stat tiles, cards).

What changes, rule by rule

PART	ON A TABLET
Targets	44 pt on iPad, 48 dp on Android, at least 8 px apart. Buttons already grow by 8 px under a finger (small 32 to 40, medium 40 to 44).
Hover	There is none. Whatever shows on hover (···, a preview, a grip) is always visible or reached by a tap; the same menu opens on a long press as on a right-click.
The reader	The best place for it. A tap on a word opens the card as a sheet that stops at half the screen, so the sentence stays visible; swipe turns pages; two columns in landscape at 19 px, one in portrait.
Drawers	A sheet from the right in landscape (440 px), from the bottom in portrait, with a handle; both rise above the on-screen keyboard.
Keyboards	With a hardware keyboard every desktop shortcut works. Without one, none is required: each has a button.
Pencil	Not special: it is a pointer. Handwriting into a text box is the system's job; never build a canvas for it.

Room by room on a tablet

ROOM	THE TABLET'S VERSION
Home	One column of the same cards; the week and "Continue learning" as wide rows that swipe; the greeting stays big.
Cursus	The adventure trail fills the width; a lesson is a stage with Check and Continue in a foot bar the thumb can reach, in the lower third.
Codex	Reading, full screen, with the chrome hidden after a moment; the library as a grid of 4 to 6 covers.
Thesaurus · Ludus	Cards big and centred with the four answers as large buttons; Ludus games use the whole width for pairs and word banks.
Sonus · Visio	The player first, with lines in time under it in landscape or beside it in portrait; the lock-screen and control-centre controls work.
Mentis	Mundus and Things to try as a single column; Diarium with the keyboard up and the toolbar above it.
Ratio	Tiles four across in landscape, two in portrait; charts get a tap in place of hover, showing the value in a small bubble.
Settings	The groups as a list on the left and their contents on the right, as on a desktop; one column in portrait.

Installing and living with it

- **iPad:** Safari › Share › Add to Home Screen. The page asks for it once, kindly, after the first lesson, with a drawing of the button. A home-screen app runs full screen with the status bar in the page's colour.
- **Android tablet:** Chrome's own Install prompt, offered from Settings and once after a session.
- **Storage:** ask the browser to keep the data (`storage.persist()`), show how much is used, and remind about a backup file; Safari may clear a site's data after a week unused unless it is installed.
- **Audio:** playback must start from a tap; the lock screen shows title, cover and the skip keys.

DO

- Test on a real iPad in both orientations, and in Split View at one-third width.
- Keep every action thumb-reachable at the sides and foot.

DON'T

- Stretch a phone screen across a tablet.
- Put a primary action in the top-left, far from the thumb.

Touch and gestures GUIDANCE ONLY

GESTURE	IN LINGUA	DESKTOP TWIN
Tap	Does what a click does, everywhere. Targets grow to 40 or 44 px under a finger.	Click
Press a word and hold (430 ms), then lift	Opens the word's drawer in the reader, with a small tap on Android.	Right-click, or D
Press a word and slide	Chooses a phrase and shows its bubble.	Drag over words
Scroll, flick	The page's own scrolling; nothing of ours intercepts it.	Wheel
Pinch	The browser's zoom. The reader's text size is its own panel.	Ctrl + wheel
Swipe a card away	Not built. PROPOSED for reviews: right for Good, left for Again, with the buttons kept.	Space, 1 to 4
Press and hold a list row	Not built. PROPOSED: the ··· menu, as right-click gives it.	Right-click

How a screen adapts by width

WINDOW	SHELL	WHAT A SCREEN DOES
1280 and up	Sidebar 216, column up to 1152	The desktop design, as drawn in the room pages.
1024 to 1279	Sidebar, column fills	Two columns stay; a third rail folds under the main one.
768 to 1023	Sidebar, folded to 64 if wanted	One main column; the rail becomes a row of chips or a tab; drawers keep their 440.
Below 768	Top bar and floating tab bar	One column, as in the phone chapter. Drawers rise as sheets.

The first row is built; the others are how it should behave, and need a real device to prove.

The iPad, before calling a screen done

- ☐ Installed from Safari: Share › Add to Home Screen. It opens without browser bars (`display: standalone`).
- ☐ Both orientations; the app asks for no lock (`orientation: any`).
- ☐ Split View and Stage Manager at a third of the screen: below 768 px it changes to the phone shell by itself.
- ☐ With a trackpad: hover states appear; with a keyboard: the shortcuts, Alt as Option, with the Mac symbols chosen in Settings.
- ☐ Safe areas: nothing sits under the home indicator (the page uses `viewport-fit=cover`).
- ☐ Sound starts only after a tap; Visio and Sonus wait for one. No vibration on iOS: nothing may depend on it.
- ☐ Text is at least 16 px in a field, or Safari zooms in on focus (the inputs already do this on a coarse pointer).
- ☐ Storage: installed to the Home Screen, the data is kept; in a plain tab Safari may clear a site that hasn't been opened for seven days. A backup reminder helps.
- ☐ The keyboard, when open, doesn't cover the field or the primary button.
- ☐ Checked on a real iPad, not only in a narrowed desktop window.

Android tablets

Chrome offers to install the app from its menu. Gesture navigation and the three buttons both leave the bottom edge to the system, so the tab bar floats clear of it (a 54 px target, glass behind). Pick the voices on the device; they differ by maker. Check on a real Samsung and a Pixel tablet: their window and keyboard behaviours are not the same.

CHAPTER 29

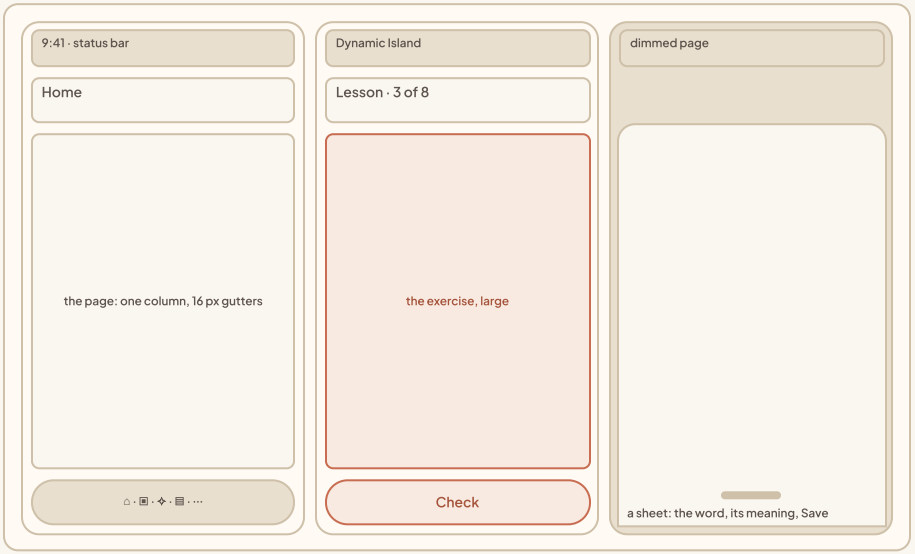
Phone: iOS and Android

GUIDANCE ONLY The phone is for the minutes in between: a lesson on the bus, ten cards in a queue, a song on a walk. Small, one-handed, interrupted. The design is the same book with fewer things on each page.

Where it stands. A phone shell exists (below 768 px) and works as it is; it hasn't had the care the desktop has had, and it isn't the focus. This chapter is the guide for the day it is.

WHERE

`src/app/shells/touch`: `TouchShell`, `TopBar`, `TabBar`.



1. **A page:** a top bar of 48 px whose big title collapses as you scroll; one column; a floating tab bar.
2. **A stage:** the tab bar steps aside; a foot bar with the one thing, in the thumb's reach.
3. **A sheet:** what was a drawer or a popover, from the bottom, with a handle, over a dimmed page.

The measures

PART	SIZE
Phones, in CSS pixels	360 to 430 wide; 640 to 932 tall (a small iPhone is 375 × 667)
Top bar	48 px below the status bar; the title shows in it once the big one has scrolled away
Tab bar	a floating rounded bar, up to 448 px wide, items 54 px high, with the finger's press shrinking an item to 94%
Side gutters	16 px, plus the safe area in landscape
Targets	44 pt (iOS) or 48 dp (Android); 8 px between
Text in a field	16 px at least, or iOS zooms the page

Layout, rule by rule

PART	ON A PHONE
Columns	One. Two panels stack; a table becomes a list of cards with the main fact large and the rest as a line; a rail becomes a Filters button opening a sheet.
The one thing	In the foot bar, above the tab bar, full width, in the thumb's reach; the title row keeps only a quiet action.
Room tabs	A scrolling row of chips under the title, never wrapping to a second line; the current one scrolled into view.
Lists	Rows 56 px or more; a swipe left reveals the ··· actions; a long press opens the same menu as a right-click.
Sheets	From the bottom; a handle; they stop at half height by default, drag up to full, drag down to close; the button at the foot clears the home indicator.
The keyboard	The focused field and its button stay visible above it; the page scrolls rather than the sheet jumping.
Toasts	Above the tab bar (about 100 px up), with the safe area, never under the Dynamic Island.
Type	The language 19 to 20 px; body 15 to 16; nothing under 12.5. Respect the system's text size up to 130%.
Motion	The same springs, with a swipe back from the left edge on iOS and the system back on Android, both returning to the exact place.

Safe areas and orientation

- The page covers the whole screen (`viewport-fit=cover`) and keeps clear of notches and home bars with `env(safe-area-inset-*)`: top for the status bar and Dynamic Island, bottom for the home indicator or Android's gesture bar, left and right in landscape.
- Both orientations work. In landscape a lesson uses two columns, the exercise and its help, and the tab bar hides.
- The status bar takes the page's colour; light page, dark text; Nocturne the other way.

iOS and Android differ here

	IPHONE	ANDROID
Back	Swipe from the left edge; a back link in the bar.	The system back button or gesture, always handled.
Install	Safari › Share › Add to Home Screen; no automatic prompt, so a drawing explains it.	Chrome offers "Install app"; Lingua can ask at a good moment.
Icons	A 180 px touch icon, light and dark.	A maskable icon, safe within the central 80%.
Notifications	Only for an installed home-screen app, and only after a tap.	After a tap; a quiet channel for reminders.
Sound	Starts on a tap; the lock screen shows controls; silent switch respected.	As iOS, with media controls in the shade.
Files	Picker through the share sheet; downloads are saved to Files.	The system picker; downloads to the Downloads folder.

Rooms on a phone: what stays and what folds

ROOM	STAYS	FOLDS AWAY OR WAITS
Home	Greeting, today's minutes, Next up	The week, first steps, Mundus's card below the fold
Cursus	The next lesson, the trail, the check	The full unit map; grammar and alphabets under Tools
Codex	Reading: the text, a tap for a word, page turns	The library's filters, import, the end-page statistics
Thesaurus · Ludus	Review the due cards; one game	Browsing decks, the dictionary, imports
Sonus · Visio	Play, pause, the current line, the next line	The library, lyrics editing, the extension's setup
Mentis	Mundus's three steps, a short Things to try	Sermo's scene list; long writing (Diarium works with the keyboard)
Ratio	Four tiles and the heat map	Every other panel, behind a tab; Tabula as cards
Settings	Look and feel, listening, your data	The rest, one group at a time

Rules for hands

DO

- Put what is used most in the thumb's arc: the lower two-thirds, toward the dominant side.
- Make an exercise answerable with one thumb.
- Save a half-finished lesson silently when the app goes to the background.
- Allow interruption: a call, a notification, the screen locking.
- Work on a bad connection, and say when it isn't.

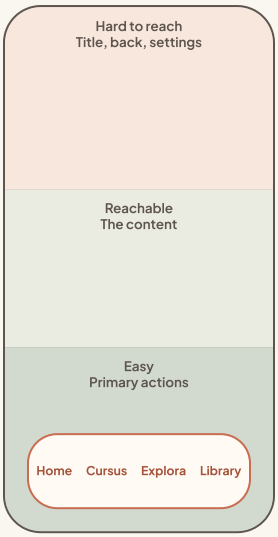
DON'T

- Put a close button top-left of a sheet and nothing else; a swipe must close it too.
- Rely on a swipe alone; each has a button.
- Auto-play sound on opening a page.
- Ask for a permission at the first start: ask where it's needed.
- Show a banner over the work to ask for a rating.

Check it on a phone

- A small iPhone (375 × 667) and a large Android (412 × 915), both orientations.
- With text at 130%, in Nocturne, with Reduce Motion on, on a poor connection.
- One-handed: can the main thing be done with the thumb alone?
- After the screen locks and unlocks in the middle of it.

Where the thumb reaches GUIDANCE ONLY



One hand, right thumb, a 6" phone.

- **The bottom third is the easy third.** The floating tab bar (items 54 px high) and the primary button of a stage live there.
- **The top is for reading, not tapping:** the title and a back link; a rare action there is a feature.
- **A destructive action goes high and far from the thumb,** with Undo.
- **One column, 16 px gutters,** and a drawer becomes a sheet that rises from the bottom.
- **Fields are 16 px or more,** so iOS doesn't zoom in; the keyboard may hide half the screen, so the primary button rides above it.

Gestures and system habits

SYSTEM	WHAT THE USER EXPECTS	WHAT LINGUA DOES
iOS back swipe	From the left edge goes back.	The browser's history; don't put a horizontal scroller at the edge.
Android back	Closes a sheet, then goes back, then exits.	Esc and back close the topmost drawer first.
Pull to refresh	Reloads the page.	Lingua is local; a reload is never needed. Don't add it.
Install	iOS: Share › Add to Home Screen. Android: Chrome's menu.	It then opens standalone, its own colour in the status bar (theme-color, by scheme).
Notifications	Asked for with a reason, once.	Not used today; the reminder is Home's Next up. PROPOSED if added.
Haptics	Subtle, on Android Chrome only.	Lessons and the reader's word hold; never the only signal.

The desktop-only rule stands: none of this is built or checked for. Keep this page as the brief for the day a phone design pass is asked for.

PART FOUR

Page by page

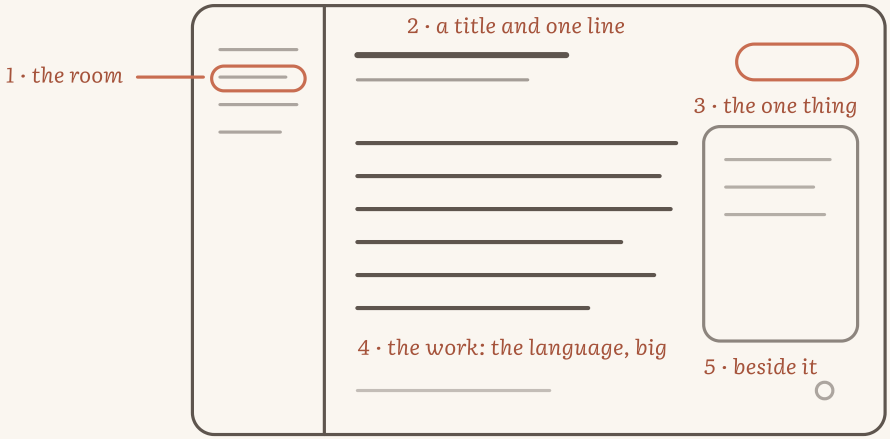
The anatomy of a page, the first start, and every room, one page each.



CHAPTER 30

The anatomy of a page

Most Lingua pages are made of the same five parts. Get these right and the page is nearly done.



USE THESE FIRST

Button, Card, Dialog, Drawer, EmptyState, Popover, SearchInput, Segmented, Select, Slider, Switch, Tooltip: `src/design/primitives`.

1. ***The room,*** *lit in the sidebar. Nothing else tells you where you are.*
2. ***A title and one line.*** *The title in Literata; the line under it in the helper's voice, and that's where a smile goes.*
3. ***The one thing to do,*** *in the accent fill, top right or under the work. One per page.*
4. ***The work.*** *The language, a list, a player: the biggest, calmest thing.*
5. ***What's beside it, and the quiet edges:*** *a panel or rail; counts, More, settings.*

CHAPTER 31 · LINGUA 7 BEGINS HERE

The first start

The first start is the first page of the book. It should feel like opening one: unhurried, a little exciting, and over before it gets long.

THE PLAN

Lingua 7-1 in `docs/LINGUA-7-PLAN.md`, from the user's notes of 2026-10-26. Code: `src/app/onboarding`.

Your name · Welcome · What Lingua can do · Language · Level · ~~You speak~~ · ~~Why~~ · Goal · Study light · Plans

Three pages to say hello, then four steps to set up, and the plans. "You speak" and "Why" were taken out: the fewer questions before the first page of reading, the better.

DOTS

Only the steps get dots, not the hello pages. Every page fits 1366 × 768, except the language page, where the list peeks in on purpose.

Your name

"First things first." One field for a first name, and Continue waits until there is one. No skip. The quiet line stays: "It stays on this device. We just like to be polite."

OUT

"Skip for now", and the Lingua mark above the title: the header already has the wordmark.

The language

The world first, wide. Click a country, and a panel opens with the country itself as its title, its flag and its name, set properly, then the languages you can learn there. No tabs. Under the map, the six core languages as small book covers, each wearing a touch of its language; under them every other language, larger than before, by tier. Each tier says plainly what it has and what it doesn't have yet.

TIERS

Core · Full · Beta · Basics · Experimental. One table, `TIERS` in `src/core/lang/languages.ts`, with *has* and *not yet*.



The level

THE CARD

Springs in on *snappy*; arrow keys move along the steps. From A2 up it says the most common words start as known.

Seven steps, A0 to C2, like a staircase. Hover one or choose it, and a small card pops up with what that level could mean: what you can read, follow and say there, and what Lingua will do about it. No example sentences, no test.



The goal

OUT

~~"B1 in about 7 months"~~, ~~"Ambitious. We like it."~~ Not accurate enough.

A level to aim for, and minutes a day. Nothing forecasts. Instead each choice says what it means: the goal in the level card's words, the minutes in what they hold in Lingua: "Cards, a few pages and a song."

Study light, and the plans

PLANS

`features/plans/PlanPitch`. Max shows monthly first.

Every palette as a small page of the app, with Follow my device first. Then the plans, which promise only what is true, with Continue free as clear as the rest.

Ask yourself

1. *Can someone reach the map in under a minute?*
2. *Does any page promise something Lingua can't measure?*
3. *Is there more than one joke on a page, or one inside an instruction?*

CHAPTER 32

The first start, screen by screen

The first four minutes decide whether Lingua gets a second day. Ten screens, each asking one thing, in a fixed order; the first three are a welcome, the last seven build the learner's plan.



Real titles and button labels, from `src/app/onboarding`; the drawings are schematic. The order is `FIRST` in `pages.ts`. The progress dots count only the seven steps from Language on ("Step 1 of 7"), so the welcome never feels like homework.

What each screen is for

#	SCREEN	WHAT IT ASKS, AND WHAT THAT IS USED FOR	PAGE WIDTH	DESIGN NOTE
1	*Name*	A first name; greets, never sent anywhere.	768	A line, not a box: a bare underline that turns terracotta on focus. Serif 40 for the title.
2	*Welcome*	Nothing. A warm line.	768	Title at 44, the quip in italics, the book last at 0.55 s. Nothing to read twice.
3	*About*	Nothing: three promises.	1152	Read anything, Keep every word, Listen and watch. No screenshots.
4	*Language*	The language being learned: picks the pack, the course, the voices.	1280	A map and five tiers; each tier's promise is on hover. The button names the language.
5	*Level*	The last line understood: seeds known words and the starting level.	768	Real text, a lines ladder; the ticks are drawn as you click.
6	*You speak*	Languages already known: what meanings are shown in.	768	Level chips A1 to C2 and native. Centred title.
7	*Why*	Reasons: orders Home's suggestions.	768	Chips with an icon; many allowed; the sentence is the learner's language.
8	*Goal*	A goal and minutes a day.	768	A default by level; the ring previews the daily target.
9	*Light*	Paper, Nocturne or the device; a palette.	1152	The language's own sample text on each, so the learner picks on reading.
10	*Plans*	Free, Lingua+ or Max.	1152	Cards for the three; Free proceeds at once.

RULE

One question a screen. A reason for each. Skippable until the language.

Everything it asks can be changed later, and every answer has a use that the learner can see: the language and level pick the starting text, the reasons order Home's suggestions, the goal sets the ring, the light sets the room. If a question wouldn't change anything, it isn't asked. A learner can replay it from the Device lab.

The frame around every screen

- **The foot:** Back as a ghost at the left, the large primary button filling the rest. Its label says what comes next (“Choose a language”, “Continue with Español”, “Begin”), not “Next”.
- **Esc** leaves a replay; on the real first start it is the learner’s to skip only once something has been chosen.
- **Motion:** steps move on the gentle spring (the slowest, 200 · 28), and the card tilts a little in perspective; with reduced motion they simply change.
- **What is saved:** the choices are written together at the end, so closing the window halfway leaves nothing half-made.

MEASURE

- Time from opening to the first card or page read.
- Where people stop: the language and level screens, first.

NEVER

- Ask for an account, an email or a card.
- Add a screen for something that could be a setting.

Watching someone do it

One learner, one sitting, no help. Say nothing, and note:

SCREEN	WHAT TO WATCH
Name	Do they type, or look for the skip?
Language	How long they scan the map; whether they read the tiers; whether a Soon language frustrates.
Level	Whether “the last line you understand” is understood; whether they hesitate over the first click.
Goal	Whether the default is accepted, raised or lowered; whether “minutes” is clear.
Light	Whether they notice the sample text, and choose by it.
Plans	Whether Free is found at once, and what they say about the others.
After	“What do you think Lingua is for?”



CHAPTER 33

Home

Your hearth

Coming home to the language: what you did today, and the one thing to do next.

SHOW FIRST	The greeting by name, today's minutes against the goal, and Next up.
KEEP QUIET	The week and its numbers, the Lingua+ button (free plan only), first steps once they're done.
THE ONE THING	The first row of Next up.

TODAY

"Good evening, {name}"; minutes in serif numerals, "8 of 15 minutes", over a ribbon split by room. Next up: at most four rows. `features/home/HomeToday.tsx`.

Home is a page you glance at, not one you study. A returning learner should be back in their book within a click, without reading a word of the interface.

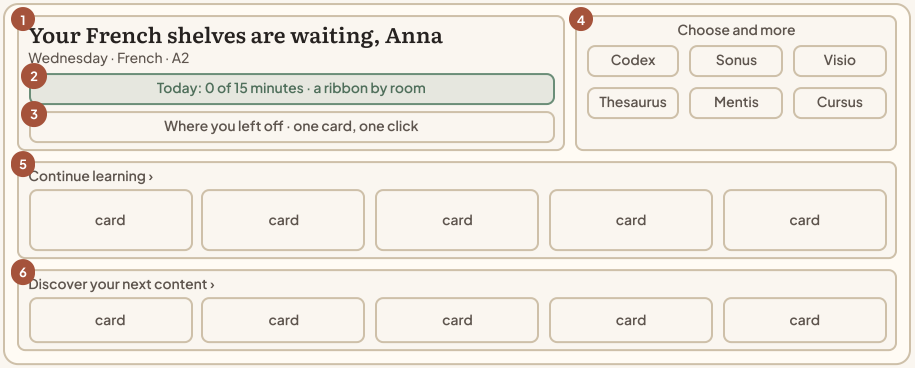
Ask yourself

1. *Can a returning learner start in one click, without reading?*
2. *Is anything below fighting Next up for attention?*
3. *Does a brand-new learner see a way in, rather than empty numbers?*

ALSO

Home layout Today / Classic in Settings › Look and feel. Both columns end on the same line.

HOME · HOW IT IS LAID OUT



1. **A greeting in the display face,** with the date and the language above it in capitals.
2. **Today,** the minutes against the goal, split by room; the only number on the page that moves.
3. **Where you left off:** the one place to carry on, as a card.
4. **Choose and more,** the other rooms as a small grid, with “Customise”.
5. **Continue learning:** a row of up to five cards, scrolling.
6. **Discover:** a row of what is new for your level; below, the week and the next grammar topic.

RULE	HOME'S VERSION
Purpose	“Where do I carry on?” Every part answers it or points to a room that does.
One thing	There is no accent button: “Where you left off” is the one thing, and the whole card is its button.
Rows	Five cards across at 1440; each card is a cover, a small label, a title, one line, and “Continue”. The row’s title is a link to its room.
Numbers	At most three: minutes today, known words, hours this week. The rest are in Ratio.
New learner	First steps replace the rows until they’re done: three tiles, each opening a room with one task.

TIPS

- Greet briefly; the second line is where a smile goes.
- Order the rows by what they’d do next, not by room.

AVOID

- A streak that shouts, or a leaderboard.
- More than two rows above the fold.



CHAPTER 34

Cursus

The course, grammar, alphabets and typing

Learning in order, for those who like a path: it leads, and everything stays open.

SHOW FIRST	The next point on the trail.
KEEP QUIET	Everything past the next few steps. Any topic can still be opened; it just isn't pushed.
THE ONE THING	Continue.

A grammar rule is easiest to keep when it comes with a sentence you have already met. Wherever Cursus can, it borrows from what the learner read or heard.

TABS

Course · Grammatica · Alphabets · Typing, keys 1–4, opening on the tab used last. The course is for the six core languages; the path became an adventure in 5.1-3.

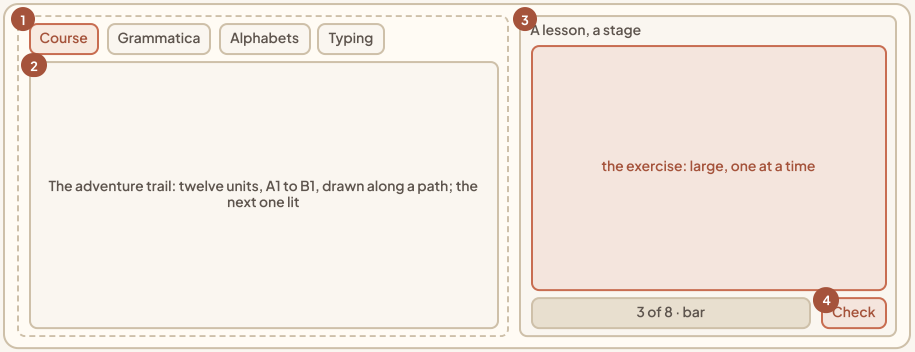
Ask yourself

1. *Is the next step obvious, and is everything else still within reach?*
2. *Does a rule show a sentence the learner has actually met?*
3. *Does a lesson end by pointing back into reading or listening?*

DECIDED

D10 of the 5.1 plan: "It leads, everything stays open."

CURSUS · HOW IT IS LAID OUT



1. **Room tabs** with a chip of progress each; the course shows only for the six core languages.
2. **The trail:** units as places on a path, the finished ones inked, the next one lit and larger.
3. **The lesson,** full width on desktop, one exercise at a time.
4. **The foot bar:** Skip at the left, Check then Continue at the right; progress above it.

TAB	LAYOUT AND FEEL
Course	The trail is the page; a unit opens a short list of its lessons in a panel at the side. The trail never asks to be scrolled sideways.
Grammatica	A guided path: “the next step, with its reason”, then a ladder of levels A1 to B2. A topic is a stage with the explanation at the left and the practice at the right.
Alphabets	A grid of letters, each a large glyph with its sound; a tap plays it, a second opens the stroke order. Only the seven scripts that need it.
Typing	The line to type, large, in the language’s keys; a keyboard map under it, the next key lit; speed and accuracy in a small row.

TIPS

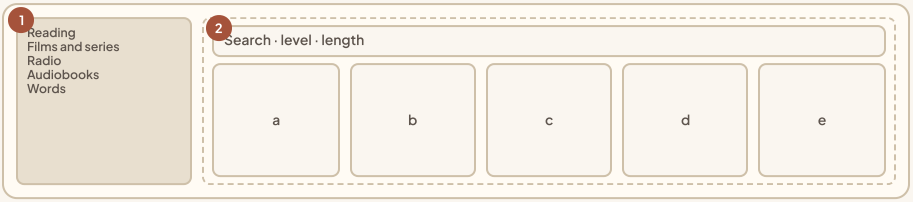
- Show how far, never how far behind.
- End every lesson on a clear next step.
- Keep the foot bar in the same place for every exercise type.

AVOID

- Lives, hearts or timers that punish.
- A lesson that scrolls: if it must, it is two exercises.

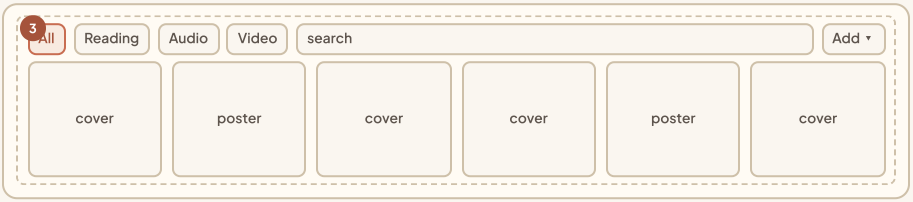
EXPLORA AND BIBLIOTHECA · HOW THEY ARE LAID OUT

Explora: one place to find something new



The kinds at the left, the finds as cards at the right, filtered to your level. The one button on a card adds it to Bibliotheca or starts it. Never a feed that goes on for ever: a page of about twenty, and More.

Bibliotheca: everything that is yours



1. **Explora's rail** picks the kind; its choice is remembered.
2. **The filters** are two at most: level and length.
3. **The kinds as tabs** and one search; **Add** (a file, a link, a camera) at the right.

PART	HOW
Covers	All the same ratio per kind (a book 2:3, a poster 2:3, a song square), so a row lines up; a drawn one where there's no cover; a thin bar for progress at its foot.
Sort	Last opened first; a popover for title, date added, length. Remembered per kind.
Selecting	A tick appears on hover; with something selected a bar rises at the foot with Tag, Move, Remove.
Opening	A click opens the thing in its own room (Codex, Sonus, Visio); the back link returns to the same scroll place.
Empty	"Nothing here yet" with Add as the one thing, and Explora as the other.

CHAPTER 35

Codex

Reading



Reading real things in the language, with every word a tap away. The heart of Lingua.

SHOW FIRST	The book you're reading. In the reader, the page and nothing else.
KEEP QUIET	Translations, grammar and the dictionary, in the word drawer, once a word is tapped.
THE ONE THING	Keep reading.

The reader is the one page that should feel like paper and nothing more. Words you are learning are marked kindly, not like mistakes; words you know are not marked at all.

MARKS

New words on `--learning-bg` with `--learning-line`; unseen ones a sage line. Styles in `src/styles/reader.css`; papers per light.

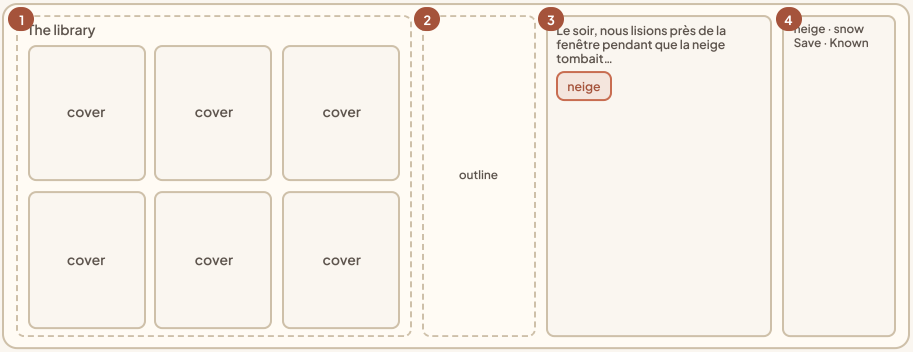
Ask yourself

1. *Is the text the biggest, calmest thing on the screen?*
2. *Does closing the drawer put you back on the very word you tapped?*
3. *Does the end of a book feel like the end of a book?*

ALSO HERE

Import a book or a comic; the end page with its ornament; the reading goal's note.

CODEX · HOW IT IS LAID OUT



1. **The library:** covers 2:3, a progress bar at the foot of each, filters and import on top.
2. **The outline,** folded until asked for.
3. **One column of text,** 640 to 720 px, the language at 19 px, a line height of 1.7.
4. **The word panel,** on a click, beside the text and never over it.

PART	HOW
The page	Paper, the reader's own: seven light or dark papers. Margins of 15% or more; chrome fades after a few seconds and returns on a mouse move.
Words	Unseen: a thin sage underline. Learning: a terracotta wash. Known: nothing. Never colour alone: the learning wash also has a soft underline.
The word card	The word large, its meaning, its form, a sound button, two buttons: Known and Save. Keys K, L, X, D act on the hovered word; right-click opens the details.
Phrases	Drag across words to translate a phrase; the sentence's translation appears in the card on S.
Turning	← → or a click at the page's edge; the page is two columns on a very wide window, one otherwise.
The end	A quiet page: time, words, what was new, one line about the next book.

TIPS

- Everything that isn't the text should leave when the reader stops looking.
- Keep the card's place fixed so the hand learns it.

AVOID

- Popups over the line being read.
- Justified text without good hyphenation.

CHAPTER 36

Thesaurus and Ludus

Your words, and games with them



Every word you keep, and the practice that brings it back just in time.

SHOW FIRST	The cards due today, and the words you met most recently.
KEEP QUIET	The scheduling: intervals, ease, forecasts. There for those who look.
THE ONE THING	Study what's due.

A word is more than its card. Wherever it shows, let it say where it was found: the book and the sentence, the song and the line.

Ask yourself

1. *Does every word show where it came from?*
2. *Is a card's answer hidden until the learner has tried?*
3. *Does practice end with a calm summary rather than a score to beat?*

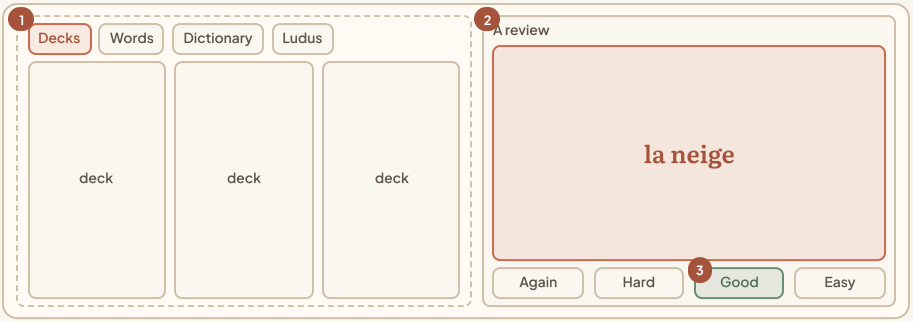
CARDS

Width Thin · Normal · Wide, per deck if wanted. Discover; sentences with a gap as a Ludus mode; the word memory and the dictionary.

CAREFUL

A saved word is not a known word. Count them separately, everywhere.

THESAURUS AND LUDUS · HOW THEY ARE LAID OUT



1. **Tabs:** Decks, the word memory, the Dictionary, and Ludus's games.
2. **The review is a stage:** the card centred, nothing else but the progress.
3. **Four answers,** large, the likeliest (Good) in sage; keys 1 to 4.

PART	HOW
Decks	Cards in folders; each shows its title, a due count in a chip, and a thin bar. Search with "/", drag to reorder, ··· for the rest.
A card	The front in the book's voice at 28 px or more; the back appears with a turn of 200 ms; the sentence it came from sits under it in soft ink.
Answers	Four buttons across the foot of the stage, equal width, with the next interval beneath each in small figures.
The Dictionary	A search at the top, the entry below: forms, recordings, the other meanings, examples from Tatoeba; the word's whole history in the word memory beside it.
Ludus	One game at a time as a stage: choices, pairs, gaps, word banks, listening, typing, speaking. The progress bar on top, Check and Continue at the foot, as in a lesson.
When done	"12 cards in 4 minutes; 3 to see again tomorrow", and the next thing.

TIPS

- Show the interval, not a grade.
- Keep Undo (Z) always one key away.

AVOID

- A countdown on a card.
- Different layouts for each game: one stage, many contents.

CHAPTER 37

Sonus et Visio

Listening and watching



Podcasts, radio, audiobooks, songs, films and shows, with the text following along.

SHOW FIRST	Continue: what's half-played or half-watched, and how much is left.
KEEP QUIET	Sources and services and their limits, until they matter. Then say them plainly.
THE ONE THING	Press play.

Only one thing plays at a time. A word looked up mid-song pauses the following, not the song, and the learner can find their line again with one click.

Ask yourself

1. *Can you look up a word without losing your place?*
2. *When a service can't do something, does the page say so instead of showing a dead button?*
3. *Is an empty library one calm card, not a wall of empty shelves?*

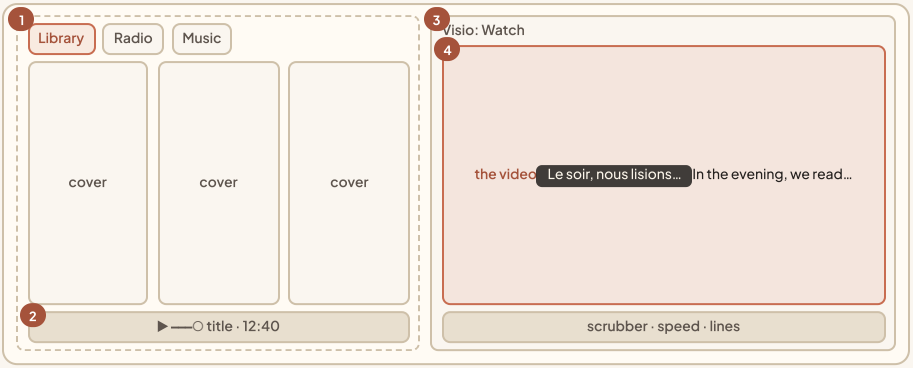
STRIP

+ Listen and + Watch; Library · Audiobooks · Radio · Streaming, and Sonus+ at the end. A sticky rail of categories in the library.

HONEST

Never imply Lingua includes someone else's subscription.

SONUS AND VISIO · HOW THEY ARE LAID OUT



1. **Sonus tabs:** Library, Radio, Music; Add at the right; a mini player stays at the foot of every page.
2. **Visio's watch page:** the video as the stage, the lines in time beneath.

PART	HOW
The lines	Two at a time, centred, 22 and 16 px, on a 78% espresso plate for contrast; every word is hoverable, and hovering pauses; ← and → step by line; , and . nudge a followed show by half a second.
The player	Play, ±10 s, speed, repeat-each-sentence (R), the transcript as a column at the right, the current line lit. ↑ and ↓ move by sentence.
A YouTube link	Plays in YouTube's own embed; its title and picture come from YouTube; the lines come from the extension or a file.
Following a show	NOT YET TRIED WITH A REAL SHOW A subtitles-only item gets a clock instead of a video: Play, Pause, ±0.5 s; the lines move in time beside the show playing in its own app.
The extension	NOT YET TRIED ON REAL SITES On Netflix, YouTube and the like it draws the same two lines over their player and asks an open Lingua tab for meanings; it reads only subtitle files, never the video.
Not built	Gemini transcripts (Max) and a YouTube captions fallback: planned.

TIPS

- Let subtitles be readable first, clever second.
- Keep the video as large as the window allows.

AVOID

- Controls over the picture that don't fade.
- Auto-play with sound on a new page.

CHAPTER 38

Mentis

The guide, talking and writing



Using the language: Mundus suggests, Sermo talks, Diarium writes.

SHOW FIRST	Today's prompt: one thing to write, one thing to talk about.
KEEP QUIET	Scores and corrections, until the learner asks for them.
THE ONE THING	Start today's prompt.

What the learner writes or says is theirs. Corrections sit beside the original, never over it, and each one can be taken or left.

AI

With the learner's own key (Groq, Gemini), and only with Max. A privacy note sits next to Check.

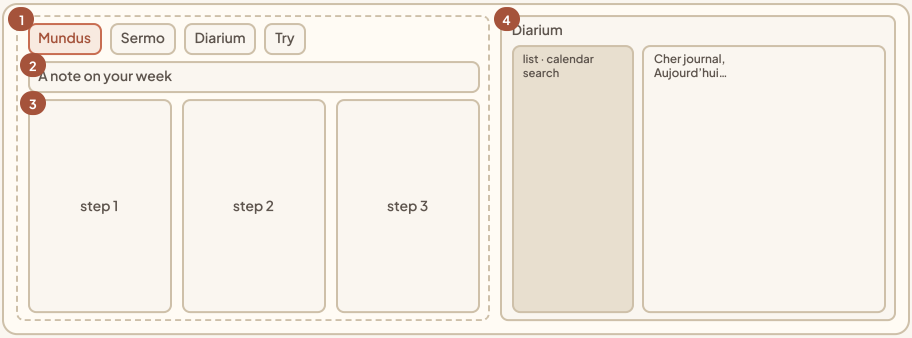
Ask yourself

1. *Are the learner's own words still visible next to the corrections?*
2. *Is it clear what is sent away, and to whom, before anything is sent?*
3. *Can a talk or an entry be left half-done and found again?*

TABS

Mundus · Sermo · Diarium, with Today's prompt on the right of each.

MENTIS · HOW IT IS LAID OUT



1. **Tabs:** Mundus (the guide), Sermo (talking with your own AI), Diarium (writing), Things to try.
2. **A note on your week,** two sentences, in the learner's language.
3. **Three next steps,** each a card that answers back: shorter, I'd rather listen, why this, not today.
4. **Diarium:** a list or a calendar at the left, the page at the right.

PART	HOW
Mundus	A note, then the three steps at the same size, one link each; "Why this" opens a line of reasoning. Nothing here changes while the page is open except when asked.
Things to try	Small cards with a time chip (5 min), a tick that stamps them, filters, and "Deal me one".
Sermo	Three screens with Back: the scene as large cards in groups, how you'll talk (voice or text, corrections as I go or at the end, 5, 10 or 20 minutes), the prompt as a reading card with Copy and the AI's buttons.
Diarium	The page is the writing: a title in the language ("Cher journal"), a plain editor, a toolbar that appears over a selection; the word count and Saved sit in the header; the AI check is a quiet line.
Calendar	Months as grids; days inked by length; a tick for corrected days; the month's totals beside it.

TIPS

- Mundus talks like a friend with a plan, not a coach.
- Let the page be quiet enough to write in.

AVOID

- A chat window inside Lingua: Sermo opens your own AI.
- Red marks all over what someone wrote.

CHAPTER 39

Ratio

How it's going



An honest look back: time, words, reading, listening, and the journey so far.

SHOW FIRST	This week, in a few numbers anyone understands.
KEEP QUIET	Every chart behind a tab or a hover.
THE ONE THING	None. Ratio is for looking; a goal can be changed from here.

Numbers are evidence, not grades. Each says what it counts and over how long. Where there isn't enough to say anything, Ratio says that kindly instead of drawing a flat line.

Ask yourself

1. *Does every number say what it counts, and since when?*
2. *Is a saved word ever called a known one?*
3. *Is any estimate labelled as one?*

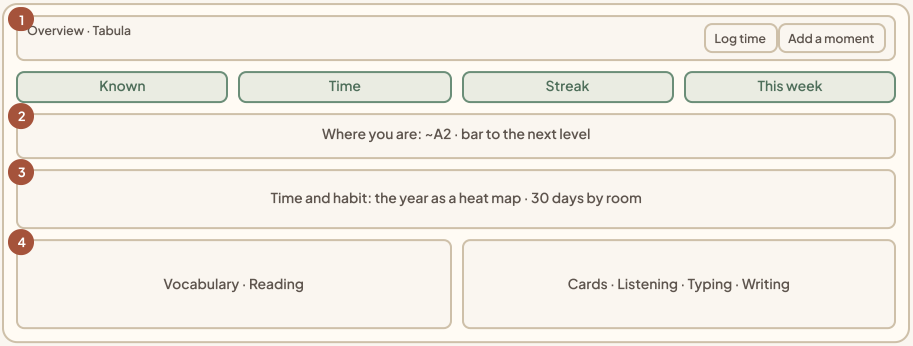
TABS

Overview · Tabula, one language at a time outside Tabula. Charts in `src/design/charts`, each with a popover.

TABULA

The languages you speak, with level and notes; certificates can be added later.

RATIO · HOW IT IS LAID OUT



1. **Tabs,** and two ghost buttons at the right: time done outside Lingua, and the moments worth keeping.
2. **Where you are,** an estimate, labelled as one.
3. **The year and the month,** which show whether you come back.
4. **Two columns of panels,** each the same anatomy: caption, numbers, a chart, a line.

PART	HOW
Language	Only the one being learned, outside Tabula. Switching language is the chip in the sidebar.
Tabula	A row per language you speak, drag to order; a slim bar for its goal and "on track", "behind" or "reached"; notes; Goal, Log time, Notion CSV and Image at the top; the reading challenge above the rows.
Panels	Vocabulary, Reading (with look-ups per 1,000 words), Listening (by kind), Typing (average and slowest letter), Cards, Writing, Grammar.
Logging	Both buttons open a drawer at the right; saving closes it with a toast, and the numbers update.

TIPS

- Say what each number counts and since when.
- Let "enough data" be a rule, not a guess.

AVOID

- Comparing with others.
- A flat line where there's no data.

CHAPTER 40

Settings and the plans

Everything you can change



Everything adjustable, in groups, with search, and the plans told straight.

SHOW FIRST	The groups, and the search.
KEEP QUIET	The fine options inside their group.
THE ONE THING	Find the setting.

Things with consequences are named for what they do. Cancelling a renewal, disconnecting a service and deleting everything are three different things, and they never share a button called Remove.

Ask yourself

1. *Can someone export, delete or manage a subscription without help?*
2. *Does each plan say its price, what it includes and when it renews, next to its button?*
3. *Is "Continue free" as easy to see as the rest?*

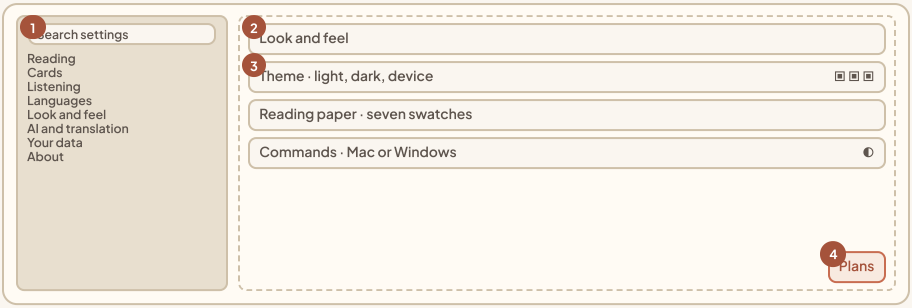
PLANS

Lingua+ €24.99 once, every update included. Max €5.99 a month or €49.99 a year: AI with your own key. Through Stripe.

NAME

Settings was called Ordo until 5.2-1.

SETTINGS · HOW IT IS LAID OUT



1. **The groups and a search,** at the left; the search finds a setting in any group.
2. **One group at a time,** with a title.
3. **Each setting is a row:** its name and one line, the control at the right.
4. **Plans** sit apart, in their own place, told straight.

PART	HOW
A row	56 px high; the name in 15 px, the line under it in soft 13 px; a switch, a segmented control, a select or a button at the right. It acts at once.
Look and feel	Choices are shown, not named: the themes and papers as swatches, the fonts in their own faces, the language touch with a preview.
Your data	Backup, restore, sync and delete are four different things with four different buttons; Delete is danger, last, and names what goes.
AI and translation	A key is pasted into a field and stays on this device; the field says so, and nothing sends it anywhere but the service it is for.
Plans	Free, Lingua+ and Max side by side, each with its price, what it includes and when it renews, beside its button; “Continue free” as easy to see as the rest.

TIPS

- Group by what a person wants to do, not by how it is built.
- Say what a setting changes right where you change it.

AVOID

- A setting that needs a restart.
- Three buttons called Remove.

CHAPTER 41

The new tabs of Lingua 6

Lingua 6 adds a few tabs to rooms that exist, and one button that appears everywhere. Each gets the same care as a room.

PLAN

`docs/LINGUA-6-PLAN.md`, the step named beside each.

TAB	SHOW FIRST	KEEP QUIET
Cursus › Sounds 6-10	The sound you're on, its drawing and recording, and Record	The full IPA charts; the waveform until you've recorded
Cursus › Exams 6-18	Your exam, the days left, the next task	Score ranges until a mock is done; past mocks
Thesaurus › Phrases 6-20	Phrases by situation, with sound	Templates and variants, until a phrase is opened
Thesaurus › Map 6-20	The word you came from, and its nearest neighbours	The whole map; filters in a rail
Mentis › Notebook 6-20	Your latest note, and a new one	Folders and tags in a rail; the error log a tab away
Ludus › Translate 6-13	The sentence to translate, large	Hints, references and the alignment view until asked
The inbox 6-2	Each word with its sentence, and three choices	Bulk actions until something is selected
Ask 6-15	A small button where the learning is	Everything: it opens a side panel, and closes back to the exact place

Ask yourself

1. *Is the new tab a place you'd go, or a feature that belongs inside another one?*
2. *Does anything generated say so, and can it be reported?*
3. *Does a word here open the same word's page as everywhere else?*

PART FIVE

Before you call it done

The states a page can be in, Lingua's check, and a sheet to tick.



CHAPTER 42

Every page's states

A page is not done when it looks right full. It is done when it also looks right empty, slow, wrong and offline.

The first time	Say why it's empty, and give one way in.
Loading	Keep the page's shape. If it takes more than a moment, say what's happening.
Nothing found	Show what was searched for, and a way to clear it.
Went wrong	Keep what the learner typed, say what failed, and offer to try again.
Offline	Say what still works. Most of Lingua does.
Not in this plan	Say what it is and what it costs, and that saying no loses nothing.
Coming	Dimmed, with a word on hover. Never a button that does nothing.
A lot	Five hundred books, ten thousand words: does it still breathe?

A toast is a whisper, not a record. Anything the learner must know stays on the page after the toast has gone.

Never show a stack trace, a key or someone else's anything.

PRIMITIVES

`EmptyState` for the first time and nothing found; `Toaster` for whispers.

KEYS

Stay on the device, travel only in request headers, never in a URL.

CHAPTER 43

The check

Lingua's own check, for every page you touch. It takes ten minutes and saves the user an evening.

FROM

`CLAUDE.md`, "How the user wants each prompt handled".

- [ ] `npm run typecheck`
- [ ] `npm test`
- [ ] `npm run build`

Then look at it

- [ ] at 1280, 1440 and a 1366 × 768 laptop
- [ ] in Paper and in Nocturne, two palettes each
- [ ] with the language touches on and off
- [ ] with reduced motion on
- [ ] with the keyboard only: can you reach everything, and see where you are?
- [ ] with a long word, a long name, Arabic from right to left, and Japanese without spaces
- [ ] at Windows' 125% scaling

Then finish it

Update the docs it touches (the roadmap, decisions, features), tick the step in the handover, give the push its version, rebase, push, and check the site went live.

VERSION

`node scripts/version.ts "…"`; Lingua 6 is `0.6.0-alpha.N`, then Lingua 7 `0.7.0-alpha.N`. See `docs/VERSIONS.md`.

CHAPTER 44

The review sheet

Page _____ Date _____

THE SPIRIT

- [ ] The language is the biggest thing
- [ ] One main thing to do
- [ ] The rest folded, not taken away
- [ ] Nothing promised that isn't built
- [ ] Nothing added behind the learner's back

THE LOOK

- [ ] Tokens only; no colour by hand
- [ ] One terracotta thing per area
- [ ] Literata for the language and titles
- [ ] Lines before shadows
- [ ] Fits 1366 × 768, no tiny scroll
- [ ] Only transform and opacity move

THE WORDS

- [ ] The title asks or names
- [ ] At most one smile, never in an instruction
- [ ] Buttons say what happens
- [ ] Plain words for money, privacy, errors

THE STATES

- [ ] The first time
- [ ] Loading keeps the shape
- [ ] Nothing found, with a way back
- [ ] An error keeps what was typed
- [ ] Offline says what works
- [ ] Coming is dimmed, not dead

THE CHECK

- [ ] Typecheck, tests, build
- [ ] Paper and Nocturne, two palettes each
- [ ] Language touches on and off
- [ ] Reduced motion
- [ ] Keyboard only
- [ ] Right to left, and no spaces

NOTES

PART SIX

The people it's for

Who Lingua is for, who it isn't for, what else they use, and how to keep listening to them.



CHAPTER 45

Who Lingua is for

Adults who learn a language by living in it: reading, listening and watching real things, at a desk, in the evening, on purpose.

SKETCHES, NOT RESEARCH

These five are the people Lingua is built for, written down so they can be tested. The private beta (Part Seven) checks them against real learners; change them when it does.

Mara, the reader

B1 IN SPANISH

WANTS	To read a whole novel, and keep the words she looks up.
TIRED OF	Apps that teach “the apple is red” for the hundredth time.
IN LINGUA	Codex, the word panel, the inbox; Discover’s classics.
PAYS FOR	Lingua+ once, for Ludus and the course’s grammar.

Tomás, the watcher

A2 IN FRENCH

WANTS	To follow series and songs without pausing every line.
TIRED OF	Juggling a subtitle add-on, a dictionary and a flashcard app.
IN LINGUA	Visio’s two lines, Sonus+ lyrics, lines saved as cards that replay their clip.
PAYS FOR	Lingua+ for Sonus+ and streaming; Max for transcripts.

Ines, the polyglot

FOUR LANGUAGES, TWO DORMANT

WANTS	To keep several languages alive and see them side by side.
TIRED OF	One app per language, and streaks that punish a week off.
IN LINGUA	Tabula, switching languages, Ratio, the radio while cooking.
PAYS FOR	Lingua+ once; it covers every language.

Jonas, the serious beginner

A0 IN RUSSIAN

WANTS	A path that leads, a new alphabet, real progress by summer.
TIRED OF	Games that feel like learning and aren't.
IN LINGUA	Alphabets, the course trail, Grammatica, typing practice.
PAYS FOR	Lingua+ for the course.

Lea, the student

B2 GOAL IN GERMAN, AN EXAM IN JUNE

WANTS	To pass the Goethe exam, and write and speak with confidence.
TIRED OF	Practice tests with no feedback.
IN LINGUA	Diarium checked, Sermo's scenes; Exams when Lingua 6 brings them.
PAYS FOR	Max during the months before the exam.

What they share: a laptop, a little time most days, a dislike of noise, and a wish to own their learning rather than rent it.

CHAPTER 46

Who it isn’t for, honestly

Saying who Lingua isn’t for keeps it good for the people it is for, and keeps the marketing honest.

Five minutes on the bus	Lingua is a desk app. Phone apps “can be” (your answer, 26 October), but not now.
Children	The voice, the content and the AI are made for adults.
Streak hunters	No hearts, no leagues, no guilt. Some people want those; they have good options.
A certificate	Lingua estimates levels; it doesn’t certify them.
A human teacher	Lingua helps between lessons. “Ask a speaker” points to people; it isn’t one.
A course in any language	The course is for the six core languages. Others read, listen and keep words.

When someone writes that Lingua isn’t for them, thank them and point them to something that is. Being useful when you lose a customer is how you keep a name.

IN THE FAQ

“Is Lingua for me?” with this list, kindly worded.

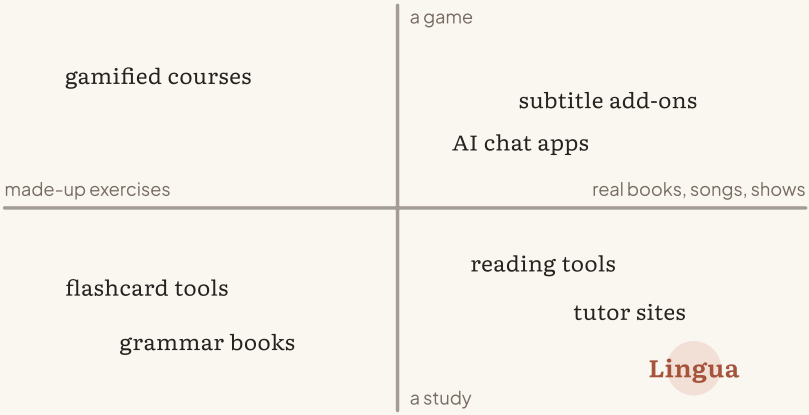
CHAPTER 47

What else they use

KINDS, NOT NAMES

Products change their features and prices often. Compare by kind here; check the real ones on the day you write about them.

Most learners use three or four tools at once. Lingua's place is the quiet desk where those tools' best parts live together.



COMPARISONS

In the EU, comparative advertising must be accurate and verifiable. Never name a competitor's weakness you can't show.

Lingua sits low and right: a study, built on real material. Its difference is not one feature but the joining up: the word met in a song is the card, the gap in a lesson and the line in your journal.

KIND	WHY LEARNERS LOVE IT	WHERE LINGUA IS DIFFERENT
Gamified courses	Easy to start, a little every day	Real material from the first weeks; no streak pressure
Flashcard tools	Powerful memory, endlessly flexible	Cards made from what you read and hear, with their sentence and sound
Reading tools	Import anything, look up everything	Bought once; also listening, watching, grammar and a course
Subtitle add-ons	Learn from the shows you watch	The same words then live in your decks and dictionary
AI chat apps	A patient partner at any hour	Lingua opens your own AI with the right scene and your words
Tutor sites	Real people, real conversation	Lingua is for the hours between lessons

Why someone would choose Lingua

- Everything in one calm place, joined by the words.
- Bought once; the free part is generous and stays free.
- Private: on your device, no account, your own AI key.
- It looks and feels like a good book, not a slot machine.

Why they wouldn't, yet

- No phone app; no friends or leagues; courses for six languages only.
- A new name from one maker: trust has to be earned in public.

CHAPTER 48

Listening to learners

Lingua collects nothing about how it's used, so learning about learners is done the old way: by asking, and by reading what they write.

Five questions for a beta learner

1. *Tell me about the last time you studied your language. What did you do?*
2. *What did you open Lingua for this week, and what happened?*
3. *Where did you get stuck or lost?*
4. *What would you miss if Lingua disappeared tomorrow?*
5. *Who do you know who should try it, and why them?*

Ask about what people did, not what they would do. "Would you pay for …" gets polite yeses; "what did you pay for last" gets the truth.

KEEP NOTES

One page per conversation in the notebook (or a folder): who, when, the five answers, one thing to change. Read them all before planning a step.

Every support email and review is a short interview. Keep a running list of what people ask for, with how many asked; it is the honest version of a feature vote.

PART SEVEN

Telling people

The story, the words, the website, where to find learners, the launch, and counting without spying.



CHAPTER 49

The story in one line

Learn a language by living in it.

USE IT

The line on the cover of the website, in the app's first page, and at the end of every post.

In a sentence: Lingua is a quiet study on your computer where you read, listen and watch real things in a language, and every word you meet is kept and comes back just before you'd forget it.

In thirty seconds

Most apps teach you made-up sentences. Lingua lets you learn from the real thing: a novel, a podcast, a song, a series. Tap any word and it's explained; keep it, and Lingua brings it back as a card, in a game, in a lesson, in your journal. For six languages there's a full course too. It runs on your computer, keeps your data there, and Lingua+ is bought once.

Four things to say again and again

Live in it	Real books, songs, radio and shows, from the first weeks.
Keep every word	One word, everywhere: met once, never lost.
A calm desk	No ads, no streak guilt, no noise. It looks like a good book.
Yours	On your device, no account, bought once, your own AI key.

The positioning, for your own notes: for adults who learn by reading, listening and watching, Lingua is the desktop study that joins every immersion tool around one record of your words, bought once and private, unlike streak games and subscription readers.

CHAPTER 50

Words for selling

Marketing speaks in the app's own voice (see "The voice"), with one more rule: every claim must be true on the day it's read.

WE MAY SAY	WE NEVER SAY
Read real books with every word a click away	Fluent in three months
Six languages with a full course	Seventy languages (alone)
Bought once, every update while Lingua is made	Lifetime access, forever, guaranteed
Your data stays on your device	100% secure
AI with your own free key, through Max	Free AI tutor; replaces teachers
Try everything free; buy when you're sure	Only today! Last chance!
Built by one person who learns languages too	A team of experts

Screenshots and videos show the app as it is, with real text from public-domain books, never a mock-up of a feature that isn't built.

Gifts are said: a creator who got a free licence says so in the review. That is the law in the EU and simple decency anywhere.

WATCH

"Lifetime" and "updates for life" are promises a buyer can hold you to. Say what you can keep.

CHAPTER 51

The website

WHERE

A page of its own on a domain of yours; the app stays at its address (today `sanchour.github.io/lingua`). The domain is your decision.

One calm page that shows the reader working, says what it costs, and gets Lingua installed in a minute.

The home page, top to bottom

1. ***The line and the reader,*** *moving: a page of a real book, a word clicked, its panel, saved.*
2. ***Three things,*** *in the fine drawings: read anything, keep every word, listen and watch.*
3. ***The rooms,*** *one line each, with a still of each.*
4. ***Languages:*** *the core six, then "and 64 more to read and listen in".*
5. ***Plans:*** *the table from "Plans and prices", as on the plans sheet; Continue free as clear as Buy.*
6. ***Private by design:*** *what stays on the device, what goes where.*
7. ***Questions:*** *is it for me, phones, my data, refunds, the AI key.*
8. ***Install:*** *one button, and the two steps for Chrome and Edge.*

ALSO

A credits page (every open source Lingua uses), the privacy policy, terms of sale, contact.

PAGES THAT FIND PEOPLE: one per core language ("Read in Spanish with every word explained") and one per way of learning ("Learn French from songs"), each true and useful on its own.

CHAPTER 52

The mark and the press kit

Make it easy to write about Lingua well: one folder with everything a writer or a creator needs, all of it true.

THE ICON

Two solid pages and a ribbon, on warm paper or warm charcoal (4.0-1). Generated with `npm run icons`.

The mark

The icon and the wordmark as the app shows them, on Cream or Charcoal, with room around them. Never stretched, never outlined, never in a colour outside the palette's ink and accent.

The press kit, a page on the website

- The one line, and a description of about a hundred words.
- The mark as SVG and PNG, light and dark.
- Six screenshots at 1440 wide, in Cream and Charcoal, with real public-domain text: the reader first, then one per room.
- A thirty-second screen recording of a word met, kept and found again.
- Prices and plans; what is free.
- A few lines about the maker, a contact, and the offer of a free licence for review.

A PICTURE FOR LINKS: 1200 × 630, the open book drawing and the line on Cream. It is what people see when someone shares the website.

CHAPTER 53

Where to find learners

For one maker with no budget: places where learners already gather, and things worth sharing there.

WHERE	HOW	CAREFUL
Learner communities	Language-learning forums and subreddits, a language's own communities and Discord servers. Share something useful (a method, a free deck), mention Lingua when it fits.	Each place's rules on self-promotion
Creators	Learners who make videos about immersion and reading. A free Lingua+ licence, no script, an honest review.	The gift disclosed
Launch sites	A launch on a product site and a "show" post on a developer forum: the local-first, bought-once, private angle.	One day, be there all day
Teachers and tutors	Lingua for the hours between lessons; a teacher's note and a free licence for their own use.	Never ask them to sell
Universities	Language centres and student societies.	Slow; worth it
A desktop store	The Microsoft Store can list a PWA (PWABuilder packages it).	An account; your decision
Word of mouth	The generous free part; a "Tell a friend" that copies a line.	No referral nagging

Not yet: paid advertising. Until the beta says who Lingua is truly for, ads would pay to find the wrong people.

CHAPTER 54

Content that teaches

The best marketing for a learning app is learning. Write and film things a learner is glad to find, whether or not they ever install Lingua.

Ideas that come from Lingua itself

- The thirty-one methods (section 4 of the Lingua 6 plan), one short guide each.
- "Your first novel in Spanish": which book, how, in what order, from a public-domain shelf.
- Songs to learn French with, by level; radio stations worth a morning.
- How a word stays: spaced repetition explained without the jargon.
- What's new in Lingua: every version, in the changelog's plain words.
- Screen recordings, a minute each: a comic's bubbles read, a song's lines saved, a book becoming a deck.

A monthly rhythm

Week 1	One guide
Week 2	One short video
Week 3	One useful thing shared in a community
Week 4	A letter to learners: what's new, one tip, one thank-you

Small and steady beats loud and rare. If a month is too full, keep only the letter.

CHAPTER 55

The launch

Three stages, each ending when it has taught what it can.

1 · Quietly (now)

VERSION
0.x, alpha.

The alpha: you and a few friends. Lingua 5 finished and checked; Lingua 6's foundations; the business set up (Part Eight). Ends when you'd hand it to a stranger without apologising.

2 · A private beta

ENDS WHEN
Most still open Lingua after three weeks, and you can say in their words why.

Twenty to fifty learners from the communities in "Where to find learners", chosen to match the five sketches. Free Lingua+ for them. A short letter each week; two conversations a week with the five questions. Fix what stops people; note what delights them.

3 · Version 1.0

VERSION
1.0.0 (VERSIONS.md: alpha ends at the first public release).

The public release, when you say so: the website, Stripe live, the launch week on the next page. Then listen, fix fast, and thank people by name.

Launch week

The week before	Website, plans and install page checked; Stripe live with a real purchase and a refund tested; support inbox and FAQ ready; the beta learners told first.
Day 1	The letter to everyone who asked; the launch-site post; be there all day to answer.
Day 2	The developer-forum post, on the local-first angle.
Day 3–4	One useful post in each learner community that welcomes it.
Day 5	Creators' reviews go live (licences sent two weeks before).
Day 6–7	Answer everything; write down every request; fix what broke.
The week after	A thank-you letter with what changed because of them.

If it goes quietly, that is normal. Most good small apps grow by being kept, not by launching loudly. Keep the monthly rhythm.



LAUNCH OFFER

If any, honest: a real end date said from the start, never a fake countdown.

CHAPTER 56

Counting without spying

DECISION

Any measuring inside the app would change the privacy promise: only with your word, opt-in, and said on the privacy page.

Lingua sends nothing about how it's used, and that is a promise worth keeping. You can still know how it's going.

NUMBER	FROM	TELLS YOU
Visitors and where from	Cookie-free site statistics (an option to choose)	Which channels work
Lingua+ sold, Max started	Stripe	Whether the free part leads somewhere
Max kept, cancelled	Stripe's portal	Whether AI is worth its price
Refunds and their reasons	Stripe, emails	Who Lingua disappointed
Licences claimed	The licence server	Paying learners, roughly
Questions and requests	The support inbox	What to build and fix
How it's going	Opt-in "How's Lingua?" once, after two weeks	The feeling, in their words

Look once a week, on the same day, in one small table. A number moving once is noise; three weeks in a row is a story.

PART EIGHT

Running it

Money, privacy and the law, support, shipping, and working with Claude across accounts.



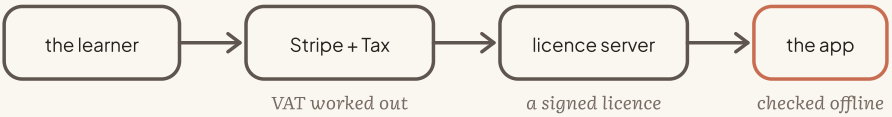
CHAPTER 57

Money

SETUP

`server/licence/README.md`, step by step, test mode first.

The learner pays on Stripe's own page; a tiny server turns the payment into a signed licence; the app checks it offline. No AI and no learning data pass through.



What has to be set up (by you)

- A business where you live, and a bank account for it.
- Stripe, with Stripe Tax on, and the EU's VAT one-stop shop (Stripe works the VAT out; the return is yours, each quarter).
- Three Payment Links, the customer portal, the key pair, the Cloudflare Worker.
- An accountant, even a small one, for the first year.

CHECK

Stripe's current fees on its pricing page before doing sums; they differ by card and country.

Costs today are close to none: GitHub Pages and Cloudflare's free tier host everything; a domain costs a few euros a year; Stripe takes a fee per payment.

What one Lingua+ leaves you

ONLY AN ILLUSTRATION

Your VAT rate, Stripe's fee and taxes on profit are yours to check.

Prices include VAT, which belongs to the buyer's country. A worked example, with the numbers as assumptions to replace:

Price paid	€24.99
VAT inside it, at 21 % (e.g. the Netherlands)	− €4.34
Net price	€20.65
Stripe's fee, assumed 1.5 % + €0.25 of the price paid	− €0.62
Left, before income tax	**about €20.03**

The sum, for any price: left = price ÷ (1 + VAT) − fee(price). Do it once for Lingua+, Max monthly and Max yearly, and keep the table beside your prices.

Refunds

YOUR DECISION

The rule, and its words in the terms of sale.

A suggestion: refund Lingua+ within fourteen days, no questions asked, and say so on the plans page. It costs little, earns trust, and buyers in the EU often have a right to withdraw anyway.

Honest limits of the licence

An app that runs on the device can be unlocked by someone determined. The licence stops casual sharing, not that. Spend no time fighting it; spend it on the people who pay.

CHAPTER 58

What Lingua could earn

NOT ADVICE

This is planning help, not financial, tax or legal advice. Check it with an accountant who knows your country.

Honestly: a good small income is a realistic first goal; a salary is a long way off. What follows are estimates from the prices and fees in this book, to plan with, not forecasts.

What one sale leaves you

PLAN	PRICE PAID	LEFT AFTER 21 % VAT AND STRIPE'S FEE
Lingua+ (once)	€24.99	about €20.03
Max, a month	€5.99	about €4.61 a month
Max, a year	€49.99	about €40.31 a year

The fee is assumed at 1.5 % + €0.25; other cards and countries cost more. VAT depends on the buyer's country. Income tax comes after.

A year, in four pictures

ASSUMPTIONS

3 % buy; 75 % Lingua+, 25 % Max yearly; no refunds, no renewals counted.

Assume that 3 in 100 active learners buy in the year (freemium apps commonly see a few per cent; yours may be lower or higher), three in four of them Lingua+ and one in four Max yearly. A buyer is then worth about **€25** to you.

ACTIVE LEARNERS	BUYERS	LEFT, BEFORE INCOME TAX
500	15	about €380
2,000	60	about €1,500
10,000	300	about €7,500
50,000	1,500	about €37,700

Read it like this: the number that matters is how many people find Lingua, not the price. Doubling the price helps less than doubling the people who try it. A one-time price also means a year's income comes from that year's new learners; Max is the part that repeats.

WHY MAX EXISTS

Its AI runs on the learner's own key, so each Max month costs Lingua nothing to serve.

What it costs to run

COST	ROUGHLY, A YEAR
Domain and hosting (GitHub Pages, Cloudflare's free tier)	€10–20
An accountant for a small business	€300–1,000, by country
A code-signing certificate, if the desktop installer needs one	€100–400
The AI subscriptions you build with	your choice; check their pages

Call it €500–1,500 a year. At €20 left per Lingua+, that is about **25 to 75 sales** to cover it, before anything is paid to you.

ESTIMATES

Every line is a range to replace with a real quote.

What moves the number

- **Reach.** The channels and content in "Telling people": a few good places beat many thin ones.
- **Conversion.** Free must show real value first; the buy button belongs beside the thing a learner just wanted. Never a nag.
- **Founders' price.** A lower Lingua+ for the first weeks (say €19.99, said plainly to be temporary) brings early cash, early reviews and early bug reports.
- **Gifts and groups.** A gift link, a teacher or class price later: more buyers from one conversation.
- **A tip jar.** "Pay what you like" through GitHub Sponsors or Ko-fi. Small, but it comes from people who love it.
- **Regional prices.** Lower prices where €24.99 is a lot; worth it once you see where learners come from. Stripe can show local prices.

What not to do

No ads and no selling learners' data: they would break the spirit of this book, the privacy promise and the licence's simplicity, for very little money. Don't raise prices on those who bought Lingua+: they keep it forever, and that promise is part of why they bought.

SEE
"Counting without spying" for how to count with no tracking.

Watch these four numbers, not more: learners who come back after a week; how many buy; how many ask for a refund; what they say when they don't buy.

CHAPTER 59

Funding it, and paying yourself

Lingua costs almost nothing to run, so the honest recommendation is to bootstrap: pay for it from what you already earn, and let sales pay it back.

NOT ADVICE

Rules of thumb for planning. Taxes, schemes and grants differ by country; ask an accountant.

Ways to fund it, in the order to try them

1. **Your own time and savings.** The costs above are small. Keep your job or studies; nothing here needs a loan.
2. **Sales.** A Founders' price and the first learners' purchases, from the launch chapter. Money in, feedback in.
3. **Small tips.** A sponsor page, mentioned once in About and once in the launch note.
4. **Grants and schemes.** Many countries and cities pay for new small businesses, software or education projects. Look up yours; they usually fund a stated piece of work, and they take weeks of paperwork.
5. **Investors: not now.** A one-time-price, local-first app is not what investors look for, and giving up a share of it would cost more than the little you need.

Keeping the money straight

- A separate business account from the first sale, so Lingua's money never mixes with yours.
- Set aside part of every payment for VAT and income tax (many people start with about a third) until your accountant gives the real share.
- Keep a runway: six months of the yearly costs, in cash, before you pay yourself anything.
- Reinvest the first small profits in what brings learners: better pages, one good video, one channel done well.

What the numbers would take

IF YOU WANT	YOU NEED, A YEAR
Costs covered	25–75 sales; about 800–2,500 learners
€5,000 left	about 200 buyers; about 6,700 learners
A €30,000 salary	about 1,200 buyers; about 40,000 learners

At €25 a buyer and 3 % buying, before income tax and the costs above.

YOUR DECISION

When to pay yourself, and how much.

A decision rule: don't quit anything for Lingua until it has paid you more than a year's costs twice over, two years running, or your savings could carry you for a year without it. Max's repeat income and a classroom price are what would change that picture; both are later.

First milestones

WHEN	WHAT TO DO
The first ten sales	Write to each buyer; ask what almost stopped them.
The first €1,000 left	Book the accountant; check the VAT and the small-business scheme in your country.
Costs covered	Put the runway aside; then spend on the one channel that works.
€5,000 left	Pay yourself a first small amount, or buy time: a month for Lingua 6's big steps.

CHAPTER 60

Privacy, licences and the law

What to have in writing before strangers pay. Not legal advice: have it read once by someone who knows.

EU

Lingua is sold from the EU, so the GDPR, consumer law and VAT rules apply.

Privacy policy	What stays on the device (all learning data); what goes where (Stripe for payments; the licence server: the checkout and the email; AI and translation providers with the learner's own key; look-ups and media from the open services Lingua names); no analytics, no ads, no selling.
Terms of sale	What Lingua+ and Max are, how Max renews and is cancelled, refunds, what "updates" means, the licence's limits.
Business details	Name, address and registration on the website, as EU law expects of a seller.
The name	"Lingua" is a common word used by many apps: search the EU's trademark register (TMview) before printing it on anything, and consider a distinctive mark.
Open content	Credits and licences for every source (Wiktionary and Lingua Libre CC BY-SA, Tatoeba CC BY, fonts under the OFL, Piper's voices each with its own licence, CLDR, LibriVox and Gutenberg's terms); Settings › About lists them: keep it complete.
Accessibility	The EU's accessibility rules exempt the smallest businesses for services, but step 6-25 is worth doing anyway.
Keys	API keys stay on the device and travel only in headers (CLAUDE.md); say so on the privacy page.

CHAPTER 61

Support

One inbox, answered by a person, in Lingua's voice. Promise a reply within two working days, and keep it.

The questions that will come

- Installing on Windows and Mac; why there's no phone app (yet).
- Moving to a new computer: device pairing, or a backup file.
- A licence lost: Settings › Lingua+ › Restore, with the licence key.
- Max and the AI key: where to get a free key, why Lingua asks for one.
- Refunds, and cancelling Max in the customer portal.
- A word with the wrong meaning; a station that doesn't play.

Each answered twice becomes a line in the FAQ. Each bug gets a note in the known-issues list on the website until it's fixed.

WORTH ADDING

A "Report a problem" that opens an email with the version and the room filled in (the Lab's reports already know them).

A reply, for example

Thanks for writing, and sorry the radio went quiet. That station changed its address yesterday; the fix is in version 0.6.0-alpha.12, which your Lingua will install by itself the next time it's closed. Happy listening.

CHAPTER 62

Shipping

Every push to main is a release: GitHub Pages builds it, and installed apps update themselves at a quiet moment, never while someone is studying.

WHERE

`docs/DEPLOY.md`, `docs/VERSIONS.md`, `.github/workflows/deploy.yml`.

Every push

1. *The check: typecheck, tests, build; a look at desktop sizes, light and dark.*
2. *The docs it touches: roadmap, decisions, features; the step ticked in the handover.*
3. *`node scripts/version.ts "what this push is"`: the version and the changelog line.*
4. *Commit as sanchour; rebase onto origin's main (other accounts push all the time).*
5. *Tag the version; push main and the tag; check the deploy went live.*

The order of the versions

0.5.2-alpha.N	Lingua 5.2 being built; 0.5.2 at its check
0.6.0-alpha.N	Lingua 6 · Features, next (your answer); 0.6.0 at its check
0.7.0-alpha.N	Lingua 7 · Design, after it
1.0.0	The first public release, when you say so

CHAPTER 63

How Lingua is built

READ MORE

`docs/ARCHITECTURE.md`; the sixty-frames rules in `docs/PERFORMANCE.md`.

A web app that installs on the desktop and works offline. Everything runs on the learner's computer; the website only hands out the app and its language packs.

Vite 8 · React 19 · TypeScript · Tailwind 4 · Dexie on IndexedDB · Zustand · motion/react · Web Workers

app/	the shells, the router, the first start
features/	one folder per room
design/	tokens, motion, fonts, the primitives
state/	short-lived state, never the source of truth
services/	side effects: storage, packs, translation, AI, transcription, audio
data/	the database, settings, backups
core/	plain TypeScript, tested: text, languages, scheduling, diff, stats
workers/	heavy work off the main thread

WHAT GOES ONLINE

Packs from the site; misses through MyMemory and Lingva; AI with the learner's key; the media services Lingua names.

Dependencies point downward only, and rooms never touch the tables directly. A tapped word is answered from the device in about a millisecond; only a word the pack doesn't know goes online, and is kept once found.

CHAPTER 64

Keeping learners' data safe

Everything a learner has done lives in their browser. Losing it would be the worst bug Lingua could have.

The database changes only by a new version with its migration, never by editing an old one; each is tried on a copy of real data first.

Backups are files the learner makes and keeps. Restoring merges or replaces; the newest change wins row by row, deletions travel along, and keys stay out unless asked.

Devices sync directly once paired, with no server between them.

The browser is asked to keep it: persistent storage is requested after the first start, so the data isn't cleared to save space.

Deleting is said plainly: removing a language names everything that goes with it; Reset says what it deletes; both ask first.

Never a migration that deletes without a backup step; never learning data sent anywhere to be stored.

NOW

Dexie version 17, `src/data/db.ts`.

WHERE

`data/backup.ts`, the merge in `core/backup/merge.ts`, both tested.

BEFORE A DATA CHANGE SHIPS

Make a backup, change, restore it, merge it: by hand once, and in the tests.

CHAPTER 65

When something breaks

Calmly, in this order: protect the learners' data, stop the harm, tell people, then fix it properly.

How bad is it?

1. **Data lost or at risk.** *Everything else waits.*
2. **Can't study:** *the app won't open, a room crashes.*
3. **Wrong language:** *a wrong meaning, a wrong rule.*
4. **Money:** *a payment or a licence that doesn't work.*
5. **Looks wrong.**

What to do

1. *Revert the push that broke it (`git revert`, a version, push). Installed apps take the fixed version at their next quiet moment.*
2. *If data is at risk, ask learners to make a backup first: a line on the website and in a letter.*
3. *Find the cause with Settings › Lab (its reports carry the version) and the tests.*
4. *Fix, check, push. Write what happened and why in DECISIONS.md, and a line in the changelog.*
5. *Thank the people who wrote in, by name.*

NOTE

A revert is a push like any other: it gets a version.

CHAPTER 66

Working with Claude

START HERE

`CLAUDE.md`, then `docs/HANDOVER.md`.

Lingua is built across several accounts and computers. Memory doesn't travel; the repository does. Everything a session needs is written down there.

CLAUDE.md	The way of working and the rules that always hold
HANDOVER.md	The queue in order, how to claim a step, the prompt to paste
The plans	Lingua 5.1, 5.2, 6 and 7, each step planned before it's built
DECISIONS.md	Why things are the way they are
This book	How every page should look, read and feel
Design notes	Settings › Lab: pin notes to anything on screen, then paste them into a session

The prompt to paste, on any account

```
Continue Lingua. Read CLAUDE.md and docs/HANDOVER.md, take the first free step in
the queue (claim it first), and do it the way CLAUDE.md describes: plan it, build
it, check it at desktop sizes, update the docs, commit and push to main, check the
deploy, and give me end notes.
```

Sessions never sign in to your accounts, create accounts, put a key in a URL, or touch anything outside the repository. Those are yours (Part Nine).

PART NINE

Start to finish

The whole road from here to a public Lingua and beyond, what only you can do, and the two checklists that matter most.



CHAPTER 67

The whole road

Ten stations. The first three are behind you; the rest is in order, and some of it can run side by side.

1 · The first builds
Lingua 1 to 3: the reader, cards, rooms, twenty languages, Grammatica and alphabets.

2 · Lingua 4.0
Tiers and seventy languages, the full dictionary, comics, the end page.

3 · Lingua 5.1
Plans and Stripe in test mode, the first start, Home, the course, typing, natural voices. Version 0.5.1.

4 · Lingua 5.2
The rooms still open (Visio, Diarium, Sermo, Mundus, Grammatica's path, Ratio), then its check. Version 0.5.2.

5 · Lingua 6 · Features
Foundations first, then the skills, paths, more of the language, around the app. Version 0.6.0. Next, by your choice.

AS OF
27 October 2026, version 0.5.2-alpha.4.

6 · Lingua 7 · Design
Room by room from your notes, starting with the first start, with this book as the guide. Version 0.7.0.

7 · Ready to sell
Your part, and it can run beside 5 and 6: the business, Stripe live, the licence server, a domain, the website, the legal texts, a support inbox (Part Eight).

8 · A private beta
Twenty to fifty learners, three weeks or more, the five questions (chapter "Listening to learners").

9 · Version 1.0
The public release and its launch week, when you say so.

10 · Growing
The monthly rhythm; what learners ask for; the open questions (sharing, phone apps) decided when the time is right.

YOUR CALL

The order of 7, 6 and the beta is yours; the handover follows it.

A suggestion to weigh: the beta doesn't have to wait for all of Lingua 6. After its foundations and the skills (Parts A and B), Lingua is more than enough to hand to strangers, and their answers are the best guide to the order of everything after.

CHAPTER 68

What only you can do

Sessions build, check and write; they never sign in, create accounts or decide for you. These are yours.

ACCOUNTS

- [ ] Stripe, with Stripe Tax and the EU's VAT one-stop shop
- [ ] Cloudflare, for the licence server (free tier)
- [ ] A domain for the website
- [ ] The Chrome Web Store, for the browser extension (5.2-4)
- [ ] Optional: the Microsoft Store, to list Lingua there

THE BUSINESS

- [ ] Register it where you live; a bank account; an accountant
- [ ] A trademark search for the name
- [ ] The privacy policy and terms of sale, read by someone who knows
- [ ] A support inbox, and the refund rule

DECISIONS STILL OPEN

- [ ] Sharing and community: files and links, or a server (Lingua 6, Q2)
- [ ] Phone apps and widgets: "can be", but when (Q6)
- [ ] The three "remove" notes read in Lingua 7's plan (its Q1–Q3)
- [ ] When the beta starts, and when 1.0 is

CHAPTER 69

Choosing what's next

There will always be more to build than time to build it. Six questions, in order, settle most choices.

1. Does it protect *what learners already have: their data, their trust*?
2. Does it stop someone *from studying*?
3. Did *several learners ask for it, in their own words*?
4. Does it *carry one word further: the same word in more places*?
5. *What is the smallest version still worth having*?
6. Could it *be described honestly on the website the day it ships*?

IDEAS

The Lingua 6 audit is an option library, not a to-do list. Only planned steps join the handover's queue.

Keep four short lists in the roadmap: now, next, later, and not. "Not" matters most: it is where good ideas go so they stop asking.

When two things tie, choose the one a learner would notice tomorrow.

CHAPTER 70

Your own rhythm

Lingua is made by one person. A book about everything would be missing a page without one for them.

- A steady week beats a heroic one: fixed hours to build, one to answer people, one to write.
- Stop at a pushed version, never in the middle of a change: the handover makes stopping safe, across usage limits and accounts.
- Keep a done list as well as a to-do list. The changelog is one; read it when the road feels long.
- Learn your own language in Lingua every day. The best tester is a learner.
- When you're away, Lingua keeps working without you; say so in the letter.
- When stuck, ship something small, or talk to one learner.



CHAPTER 71

Two checklists

BEFORE THE BETA

- [ ] Lingua 5's check passed (0.5.2)
- [ ] The first start works for a stranger, start to finish
- [ ] Backups and moving to a new computer tested
- [ ] "Report a problem" and an inbox that's read
- [ ] Known issues written down
- [ ] Free licences ready for the beta learners
- [ ] A short welcome letter and the five questions
- [ ] The privacy page says what Lingua sends where
- [ ] The credits page is complete

BEFORE 1.0

- [ ] The beta's stoppers fixed
- [ ] Stripe live: a real purchase and a refund tested
- [ ] The licence server live; test mode off by itself
- [ ] The website, plans, install and FAQ pages
- [ ] Terms, privacy, business details online
- [ ] The name checked; the domain live
- [ ] Every screen through the review sheet
- [ ] Launch week written down, day by day
- [ ] `version.ts --line 1.0.0` when you say so

And then: thank the people who helped, by name.



APPENDIX A

The numbers

Every value in one place, from `src/design/tokens.css`. The palettes change some of them; see `palettes.ts`.

TOKEN	PAPER	NOCTURNE	USED FOR
--bg	#faf6f0	#1f1c1a	the page
--surface	#efe7d9	#2b2724	cards, fields
--raised	#fffbf4	#35302c	things that float
--sunken	#e8dfcf	#191715	wells, tracks
--ink	#262220	#ebe0cc	text
--ink-soft	#5e554e	#bdb19f	second text, drawings
--ink-faint	#776c63	#918779	labels, hints
--line	#e1d6c4	#3b3532	hairlines
--line-strong	#cec0a9	#514944	field edges
--accent	#c86d51	#d49b54	the bookmark
--accent-text	#a5533b	#dba765	text in the accent
--sage	#6b8e78	#93b39e	known, done
--sage-text	#4f6f5b	#93b39e	text in sage
--inverse	#3b302a	#e8d8bb	toasts, tooltips
--on-inverse	#f8efe0	#2b231e	their text
--learning-bg	accent at 14%	accent at 17%	a word being learned
--scrim	ink at 28%	near-black at 45%	behind dialogs

Type

Serif	Literata Variable, Noto Serif Variable, Georgia	`--font-serif`
Sans	Plus Jakarta Sans Variable, Inter Variable, system-ui	`--font-sans`
Display	the language's heading face, else Literata	`--font-display`
Sizes	44 · 40 · 36 · 30 · 22 · 19 · 15 · 13.5 · 12.5 · 11.5 px	

Corners and light

Corners	8 · 12 · 16 (buttons by size, small panels) · 20 (side panels) · 24 (cards, dialogs, the map) · full (pills)
elev-1	0 1 2 ink/6%, 0 2 8 ink/5%
elev-2	0 2 6 ink/7%, 0 12 32 ink/10%
elev-3	0 8 24 ink/11%, 0 28 72 ink/17%

Motion

snappy	560 · 42 · mass 0.8	soft	320 · 34
sheet	400 · 40 · mass 0.9	gentle	200 · 28
fluid	210 · 26 · mass 1	fade	0.16 s, ease 0.2 0 0 1
page	y 8 → 0 on soft	full	y 18, scale 0.985, on fluid

Screens and keys

Check at	1280 · 1440 · 1366 × 768, light and dark
Rooms	Alt 1–8, in the sidebar's order
Cursus tabs	1–4
Contrast	4.5 : 1 for text; accent and sage text use their `-text` tokens

APPENDIX A · ON ONE PAGE

The cheat sheet

Every number here comes from one block of facts at the end of this file, and a test (`handbook-facts.test.ts`) checks that block against the code, so the book and the app can't drift apart without a failing test.

Sizes (px)

Button small · medium · large	32 · 40 · 48
Corners on those buttons	8 · 12 · 16
Drawer width (or the screen less 24)	440
Dialog, at the most	448
Sidebar, open · folded	216 · 64
Content column	1152
Text, smallest · body · reading	12.5 · 15 · 19
Check at (px)	1280 · 1440 · 1366 × 768

Time

Fade (s)	0.16
Spring snappy (stiffness · damping · mass)	560 · 42 · 0.8
soft (pages, cards)	320 · 34
sheet (drawers)	400 · 40 · 0.9
gentle (onboarding)	200 · 28
fluid (full motion)	210 · 26 · 1

Colour, Paper · Nocturne

Page	#faf6f0 · #1f1c1a
Surface	#efe7d9 · #2b2724
Ink	#262220 · #ebe0cc
Accent	#c86d51 · #d49b54
Sage	#6b8e78 · #93b39e

Keys (Alt, or ⌥ on a Mac)

Alt 1 – 8	The rooms, in the sidebar's order
[	Collapse or expand the sidebar
?	Show keyboard shortcuts
/	Search books or decks
Esc	Close a drawer or dialog
K	Mark the hovered word known
L	Mark the hovered word learning
X	Reset the hovered word to unseen
D	Open the hovered word's details
Z	Undo the last answer
E	Edit the card

Always

- One accent button a page; the rest ghosts.
- Tokens, never a hex; no pure black or white.
- Text 4.5 : 1; state never by colour alone.
- Every icon has a name; every chart a label.
- Empty, loading, error, offline: all four designed.
- Paper and Nocturne, and 1366 × 768.
- Sound only for a judged answer; off means silent.

Where things are

- Tokens: `src/design/tokens.css`
- Primitives: `src/design/primitives`
- Motion: `src/design/motion.ts`
- Charts: `src/design/charts`
- Papers: `src/design/papers.ts`
- Shortcuts: `src/app/ShortcutsDialog.tsx`

APPENDIX B

Words Lingua uses

Cursus	a course, a running. The course, grammar, alphabets, typing.
Codex	a bound book. Reading.
Thesaurus	a treasury. Your words and decks.
Ludus	a game, and a school. The games with your words.
Sonus et Visio	sound and sight. Listening and watching.
Mentis	of the mind. Mundus (the world: the guide), Sermo (talk), Diarium (a daily record).
Grammatica	grammar, in Cursus.
Ratio	a reckoning. Overview, and Tabula (a writing tablet: the languages you speak, each with its goal).
Ordo	order. Settings' old name.
Paper, Nocturne	the light and the dark.
Palette, paper	a colour set for each light; a page colour for reading.
Language touch	a flavour: a heading face, an ornament, a lean of the accent.
Tiers	Core, Full, Beta, Basics, Experimental: how much Lingua has for a language.
Lingua+	everything, once, €24.99.
Max	the AI features, with your own key, €5.99 a month or €49.99 a year.
Next up	Home's list of one-click ways back in.
The word drawer	what opens when you tap a word.
The inbox	where words wait before they become cards (Lingua 6).
The word record	one record per word: the language's half and yours (6-1).
The learner's record	the evidence of what you know and where you slip (6-5).
Ask	the tutor, a button where the learning is (6-15).
Lingyu	the design package's temporary name; its catalogue fed Lingua 6.
A0 to C2	the levels; A0 is "starting out".

APPENDIX C

Lingua 6 and 6 at a glance

Lingua 6 · Features first, then Lingua 7 · Design (your answer, 26 October). Each step is planned in detail before it is built, and checked against this book.

PLANS

`docs/LINGUA-6-PLAN.md` (step 23), `docs/LINGUA-7-PLAN.md` (step 24); the queue in `HANDOVER.md`.

LINGUA 6	STEPS	WHERE IT STANDS
Audit	1,618 entries checked; the plan written	*done*
A	6-1 word record · 6-2 inbox and decks · 6-3 cards and memory · 6-4 exercise engine · 6-5 learner's record · 6-6 checked generation	*first*
B	6-7 grammar · 6-8 reading · 6-9 listening · 6-10 sounds · 6-11 speaking · 6-12 writing · 6-13 translating	*after A*
C	6-14 courses · 6-15 the tutor · 6-16 adaptive · 6-17 goals and progress · 6-18 exams	*after B*
D	6-19 culture and the interface in the language · 6-20 phrasebook, notebook, map · 6-21 games and projects · 6-22 dictionary tools · 6-23 each language's tools	*after A*
E	6-24 import and export · 6-25 accessibility and the desktop	*any time after A*
F	6-26 experiments · 6-27 together, and beyond the desk	*your word*
G	6-28 the check; version 0.6.0	*last*

LINGUA 7	WHAT	WHERE IT STANDS
7-0	This handbook (first edition; this second edition adds Parts Two and Six to Nine)	*done*
7-0b	Design notes in the Lab: click anything in Lingua, pin a note, copy them for Claude	*done*
7-1	The first start, quieter and shorter	*planned*
7-2 – 7-9	Home, Cursus, Codex, Thesaurus and Ludus, Sonus et Visio, Mentis, Ratio, Settings and the plans	*waits for your notes*
7-10	The states, everywhere	*waits for your notes*
7-11	The check; version 0.7.0	*last*

Your answers to Lingua 6's questions

Q1	Lingua 6 first, then Lingua 7
Q2	Sharing and community: still to be considered
Q3	No live voice conversation inside Lingua
Q4	Yes: the interface in the language you learn
Q5	Yes: streaks optional, stamps not points, reminders off by default
Q6	Phone apps and widgets: can be, not ruled out

PRINT

```
node scripts/handbook-pdf.ts
```

When a step changes a rule in this book, the book changes in the same commit. Then print it again.

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FOR YOUR NOTES AND SKETCHES



Set in Literata and Plus Jakarta Sans, the faces Lingua is set in, on Lingua's own Cream. Drawn with one fine line.

LINGUA · DOCS/HANDBOOK